

Corvus Keyword Reference

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1 Introduction

Corvus is a scientific workflow framework built with python with a focus on calculations of spectroscopy. It is written to facilitate advanced calculations for novice users.

Corvus is "property driven" which means that the user specifies a property to calculate, and Corvus figures out how it can obtain that property, and what inputs are needed from the user. This is done by building a dependency tree, with the root given by the list of requested properties. The dependency tree is built by checking the dependencies of (other properties needed to calculate) the requested property, then asking if these can be satisfied by any of the available codes. This procedure is repeated until either all dependency requirements are met, in which case a workflow is built and run. If all requirements are not met, the user is notified of any necessary user input.

Most of the workflows available to standard users are meant to facilitate calculations using the FEFF10 code. These workflows include loops over specified parameters, configurational averaging, RIXS, and creation of input files from multiple available structural input such as .cif, .xyz, materials project id, or vasp output.

2 Download and Installation

2.1 Getting Corvus

To get corvus, go to <https://github.com/times-software/Corvus/releases/latest> and download the archived .zip or .tgz file.

2.2 Installation

To install corvus, you will need python. Versions > 3.12 and < 3.14 are supported. Earlier versions may work as well. We also strongly suggest that you use an environment manager such as conda to install corvus. Our suggestion is to use the free miniforge package, which can be found at <https://conda-forge.org/download/>

Python's venv environment management package can also be used.

2.2.1 Simple Installation using install script

Once you have a version of Corvus from the github website, you can follow the following simple directions to install. Note that in the below "version" should be replaced with the version that you downloaded, e.g., Corvus-3.2.0-01.zip.

MacOS Unzip the downloaded file. Open a terminal and navigate to the file. Run the install.sh script.

```
cd Downloads
unzip Corvus-version.zip
cd Corvus-version/
source install.sh
```

Windows Unzip the download file. Open a terminal and navigate to the directory. Run the installer.bat script.

```
cd Downloads
unzip Corvus-version.zip
cd Corvus-version\Corvus-version
./install.bat
```

2.2.2 Step-by-step instructions with conda

In MacOS, just open a terminal, and it will already have conda enabled if you have it installed. In windows, you will need to open a conda terminal, (Miniforge command window if you installed miniforge conda).

1. Download corvus from the github page.
2. unzip the folder.
3. In the terminal, navigate to the folder
4. Create a conda environment for corvus
5. Install corvus with pip.

```
cd Downloads/Corvus-version/
conda create -name your_corvus_env python=">=3.12,<3.14"
conda activate your_corvus_env
pip install .
```

3 Running Corvus

Corvus is run from within a terminal (or miniforge dos command window on windows). If you used the install script to setup corvus, there should be an executable script named "Corvus-version.command" or "Corvus-version.bat" on your Desktop. To run corvus, double click this script, navigate to the input file you want to run, and type,

```
run-corvus -i your_input_file
```

If you installed with the step-by-step instructions, you will need to activate your corvus environment first. With conda, this is done by typing,

```
conda activate your_corvus_env
```

4 The Corvus input file

A Corvus input file is composed of a sequence of keyword blocks.

Each keyword is followed by a brace-delimited block containing the data associated with that keyword.

Example:

```
title
{
    Example Calculation
}
```

Supported schema types:

- S = String
- I = Integer
- F = Float
- L = Logical
- P = Free-form text

5 The minimal input, and most important keywords

The starting point for any Corvus calculation is to request a property or list of properties to calculate. This is done using the keyword `target_list`. From here, corvus will look for available codes that can calculate each target, and figure out what input is needed. There are many possible targets for Corvus, but here I will list a few of the most important ones for calculating spectra using FEFF10, and how to use them. In general, when using FEFF10 an average over unique sites in the structure must be taken in order to get the full spectrum. Thus in order to perform a spectral calculation using an external structure as input (the suggested method) you must define,

```
target_list{ cfavg} # Calculate the average of some property.
cfavg_target{ xanes } # Average the XANES, could also be xes, rixs, exafs, ... or multiple properties
```

To specify the structure, you can use a variety of file formats, e.g., .xyz, .cif, XDATCAR, or you can have corvus download a structure from the Materials Project. Example:

```
cif_input{ your_structure.cif }
```

You will also need to specify which element and edge you want, e.g., for the Cu K-edge,

```
absorbing_atom_type{ Cu }
feff.edge{ K }
```

Thus a minimal input file for the Cu K-edge of CuO would look as follows:

```
target_list{ cfavg }
cfavg_target{ xanes }
cif_input{ CuO.cif }
absorbing_atom_type{ Cu }
feff.edge{ K }
```

There are of course many useful and important options available, but this will give you a calculation that has reasonable defaults. One special target to note is the "loop" target which allows the user to define loops over the values associated with any keyword or set of keywords. For single line keywords, the structure is relatively simple:

```
target_list{ loop } # Make corvus produce a loop over some parameters.
loop_target{ cfavg } # The target property to calculate for each calculation is an average
cfavg_target{ xanes } # Average the xanes over different unique sites in the structure.
```

```

# Now define the loop:
loop_parameter{
single_line_keyword
values_line1
values_line2
values_line3
...
}
# Optionally, define labels for the loop directories. Must have the same number of entries as loop_parameter
# loop_labels{
# label1
# label2
# label3
# ...
# }

```

There is also a special value for the first line after "loop_parameter" that loops over input files. This loops over any defined sets of keyword->value combinations, e.g.,

```

\begin{verbatim}
loop_parameter{
file
file1.in
file2.in
file3.in
...
}

```

This will cause corvus to perform calculations with all parameters loaded from the main input file + any parameters existing in fileN.in. Input in fileN.in overrides input in the main input file.

6 Output directory structure and output files

The output file and directory structure of Corvus is as follows: Calculations will be run in sub-directories of your current directory named Corvus_description where "description" describes what is in the directory.

Simple calculation that ask directly for a spectrum calculated by FEFF are calculated in the directory Corvus1_FEFF, while asking for an average over configurations (target_list{ cfavg }) produces a top-level directory named, Corvus3_cfavg_spectrum, where spectrum is the named spectrum that was requested by the user, e.g., Corvus3_cfavg_xanes. Then each of the subdirectories of Corvus3_cfavg_xanes, will have directories named by the index of the absorber as it was run, a label connected to which atom is the absorber, and the code (FEFF in this case).

For the special target loop, the top level folders start with CorvusN_loop_loopparameter_loopvalue. For example, if the user requests a loop over the parameter feff.corehole, with values none, rpa, fsr, then Corvus produces the directories Corvus1_loop_feff.corehole_none, Corvus1_loop_feff.corehole_rpa, Corvus1_loop_feff.corehole_fsr

Finally, the main output files of Corvus are named Corvus.target_subtarget.out. Thus for calculations that request xanes and provide the structure directly in the input file, the main output is in

Corvus.xanes.out. For configurational averages (cfavg), the main output is in Corvus.cfavg_xanes.out (for xanes calculations). Finally for loop calculations, the main output is inside the top-level loop directories, and named the same as it would be for a single calculation, e.g. Corvus.xanes.out or Corvus.cfavg_xes.out etc.

The documentation below lists every recognized keyword, its expected schema, default value, and documentation string.

7 Keyword Index by Section

Corvus Controls

- multiprocessing_level
- multiprocessing_ncpu
- scratch
- target_list
- title
- usehandlers
- usesaved
- write_input_only

General Methodology

- clusterradius
- constant_volume
- keep_symm
- method
- molecule_vacuum_margin
- mt_overlap
- nkpoints
- nqpoints
- pspfiles
- pw_encut
- scf_conv
- xc

Physical Properties

- absorbing_atom
- absorbing_atom_by_label
- absorbing_atom_type
- atomic_charge
- cell_angles_abc
- cell_scaling_abc
- cell_scaling_iso
- cell_struc_opt_flags
- cell_struc_xyz_red
- cell_vectors
- charge
- cif_input
- cluster
- fermi_shift
- ismetal
- multiplicity
- number_of_atoms
- polarization
- species
- spectral_broadening
- spin_moment
- supercell_dimensions
- vasp_outcar
- vasp_snapshot
- vasp_xdatcar
- vasp_xml
- xyz_input
- xyz_snapshot

Abinit

- `abinit.ng2qpt`
- `abinit.ngqpt`
- `abinit.verbatim`

DMDW

- `dmdw.ioflag`
- `dmdw.nlanc`
- `dmdw.paths`
- `dmdw.tempgrid`

NWChem

- `nwchem.basis`
- `nwchem.charge`
- `nwchem.iter`
- `nwchem.mult`
- `nwchem.qmd.dt_nucl`
- `nwchem.qmd.nstep_nucl`
- `nwchem.qmd.print_xyz`
- `nwchem.qmd.snapstep`
- `nwchem.qmd.targ_temp`
- `nwchem.qmd.thermostat`
- `nwchem.xas.alpha`
- `nwchem.xas.nroots`
- `nwchem.xas.vec`
- `nwchem.xas.xrayenergywin`
- `nwchem.xaselem`
- `nwchem.xc`

FEFF

- debyetemp
- dmdw_nlanczos
- feff.absolute
- feff.afolp
- feff.atoms
- feff.cfaverage
- feff.cgrid
- feff.chbroad
- feff.chsh
- feff.chwidth
- feff.cif
- feff.compton
- feff.config
- feff.control
- feff.coordinates
- feff.corehole
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- cfavg_max_configurations
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- cfavg_target

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OCEAN

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- ocean.cnbse.strength
- ocean.cnbse.xmesh
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- ocean.diemac
- ocean.ecut
- ocean.edges
- ocean.fband

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9 Corvus Controls

9.1 multiprocessing_level

Keyword	multiprocessing_level
Kind	scalar
Type	String
Default	cfavg
Deprecated	No

Which level of the calculation do you want to use multiprocessing on. Can be "cfavg" or "loop".

Schema

cfavg |X| S

9.2 multiprocessing_ncpu

Keyword	multiprocessing_ncpu
Kind	scalar
Type	Integer
Default	1
Deprecated	No

Number of processors to use in multiprocessing. This should be maximum of the number of cpus on a single node.

Schema

1 |X| I

9.3 scratch

Keyword	scratch
Kind	scalar
Type	String
Default	.

Deprecated	No
------------	----

Directory for disk scratch for those Handlers that require large amounts of disk. If the directory is not present or it can not be created, Corvus reverts to the default. NOTE: This input variable is not fully implemented yet.

Schema

. |X| S ...

9.4 target_list

Keyword	target_list
Kind	repeating-record
Type	Repeating record with columns: String
Default	Required / No default
Deprecated	No

Space separated list with all the target properties requested for this calculation.

Schema

|X| S ...

|X| ...

9.5 title

Keyword	title
Kind	repeating-record
Type	Repeating record with columns: Free-form
Default	Required / No default
Deprecated	No

Title of this calculation.

Schema

This is a Corvus calculation |X| P
|X| ...

9.6 usehandlers

Keyword	usehandlers
Kind	scalar
Type	String

Default	Required / No default
Deprecated	No

Explicitly declare what Handlers are to be used in the generation of the Workflow. This helps the current simple Workflow generator when a given target is provided by more than one Handler. Current possible values are: Feff, FeffRixs, Dmdw, Abinit, Vasp, Nwchem, Orca

Schema

|X| S ...

9.7 usesaved

Keyword	usesaved
Kind	scalar
Type	Logical
Default	False
Deprecated	No

Use previously calculated data rather than recalculating

Schema

False |X| L

9.8 write_input_only

Keyword	write_input_only
Kind	scalar
Type	Logical
Default	False
Deprecated	No

If True, write input for each of the top level calculations, then exit. Not implemented for all calculations.

Schema

False |X| L

10 General Methodology

10.1 clusterradius

Keyword	clusterradius
---------	---------------

Kind	scalar
Type	Float
Default	12.0
Deprecated	No

Radius of clusters created from cif files.

Schema

12.0 |X| F

10.2 constant_volume

Keyword	constant_volume
Kind	scalar
Type	Logical
Default	True
Deprecated	No

Keep the simulation cell volume constant in cell optimizations.

Schema

True |X| L

10.3 keep_symm

Keyword	keep_symm
Kind	scalar
Type	Logical
Default	True
Deprecated	No

Toggle the preservation of initial symmetry in optimizations.

Schema

True |X| L

10.4 method

Keyword	method
Kind	scalar
Type	String
Default	dft

Deprecated	No
------------	----

Method used to compute the target. This variable is somewhat context-dependent and is currently not fully implemented. Current possible values are: dft, mp2, ccscf

Schema

dft |X| S

10.5 molecule_vacuum_margin

Keyword	molecule_vacuum_margin
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Added to the largest distance between two atoms in the molecule before making a unit cell

Schema

|X| F

10.6 mt_overlap

Keyword	mt_overlap
Kind	repeating-record
Type	Repeating record with columns: String, Float
Default	Required / No default
Deprecated	No

set the muffin-tin overlap for different elements.

Schema

H 0.8 |X| S F
|X| ...

10.7 nkpoints

Keyword	nkpoints
Kind	vector
Type	Vector: Integer, Integer, Integer
Default	1 1 1
Deprecated	No

Define the number of k-point in each direction of the grid for reciprocal space simulations.

Schema

1 1 1 |X| I I I

10.8 nqpoints

Keyword	nqpoints
Kind	vector
Type	Vector: Integer, Integer, Integer
Default	Required / No default
Deprecated	No

Define the number of q-point perturbations in each direction of the grid for density functional perturbation theory (DFPT) simulations.

Schema

|X| I I I

10.9 pspfiles

Keyword	pspfiles
Kind	repeating-record
Type	Repeating record with columns: String, String
Default	Required / No default
Deprecated	No

List of pseudopotentials for each atom in the system. Each line contains the label for the atoms (usually the element with the pseudopotential). The required format for the files will depend on the Handler used.

Schema

|X| S S
|X| ...

10.10 pw_encut

Keyword	pw_encut
Kind	scalar
Type	Float
Default	15.0
Deprecated	No

Planwave energy cutoff for reciprocal space simulations, in au.

Schema

15.0 |X| F

10.11 scf_conv

Keyword	scf_conv
Kind	scalar
Type	Float
Default	1.0e-5
Deprecated	No

Convergence threshold for SCF cycles (HF, DFT, etc) in au. The value set by this variable might be internally overridden if the target requires tighter convergence settings.

Schema

1.0e-5|X| F

10.12 xc

Keyword	xc
Kind	scalar
Type	String
Default	lda
Deprecated	No

Exchange-correlation functional to use if the "method" selected is "dft". Current possible values are: lda, pbe, b3lyp

Schema

lda |X| S

11 Physical Properties

11.1 absorbing_atom

Keyword	absorbing_atom
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Index of the absorbing atom for core spectroscopies such as XANES, EXAFS and XES. The index counts from one for the first atom in the associated cluster (see above).

Schema

|X| I ...

11.2 absorbing_atom_by_label

Keyword	absorbing_atom_by_label
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Set the absorber by the label associated with it in the cif file. Will set all atoms that have a label that starts with this string as absorbers.

Schema

|X| S ...

11.3 absorbing_atom_type

Keyword	absorbing_atom_type
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Symbol of atom you would like to calculate the XANES for. Will loop over symmetry unique sites defined in the CIF file denoted by cif_input (see below).

Schema

|X| S ...

11.4 atomic_charge

Keyword	atomic_charge
Kind	repeating-record
Type	Repeating record with columns: String, Float
Default	Required / No default
Deprecated	No

charge to be placed on each atoms of a given species. atomic_symbol charge ...

Schema

|X| S F

|X| ...

11.5 cell_angles_abc

Keyword	cell_angles_abc
Kind	vector
Type	Vector: Float, Float, Float
Default	Required / No default
Deprecated	No

Scaling of the a, b and c axes of the simulation cell, in Angstroms.

Schema

|X| F F F

11.6 cell_scaling_abc

Keyword	cell_scaling_abc
Kind	vector
Type	Vector: Float, Float, Float
Default	Required / No default
Deprecated	No

Scaling of the a, b and c axes of the simulation cell, in Angstroms.

Schema

|X| F F F

11.7 cell_scaling_iso

Keyword	cell_scaling_iso
Kind	scalar
Type	Float
Default	1.0
Deprecated	No

Unitless isotropic scaling of the unit cell.

Schema

1.0 |X| F

11.8 cell_struc_opt_flags

Keyword	cell_struc_opt_flags
Kind	repeating-record
Type	Repeating record with columns: Logical, Logical, Logical
Default	Required / No default
Deprecated	No

Toggle the optimization of the x, y and z coordinates of each atom in the system.

Schema

```
|X| L L L  
|X| ...
```

11.9 cell_struc_xyz_red

Keyword	cell_struc_xyz_red
Kind	repeating-record
Type	Repeating record with columns: String, Float, Float, Float
Default	Required / No default
Deprecated	No

Structure of the simulation cell in reduced coordinates if the system is extended and has periodic boundary conditions.

Schema

```
|X| S F F F  
|X| ...
```

11.10 cell_vectors

Keyword	cell_vectors
Kind	matrix
Type	3 rows of Float, Float, Float
Default	Required / No default
Deprecated	No

Normalized simulation cell vector directions. These vectors are scaled using the "cell_scaling_iso" and "cell_scaling_abc" input variables to generate the simulation cell.

Schema

```
|X| F F F  
|X| F F F  
|X| F F F
```

11.11 charge

Keyword	charge
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Net charge of the system.

Schema

|X| I

11.12 cif_input

Keyword	cif_input
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

cif input file name.

Schema

|X| S

11.13 cluster

Keyword	cluster
Kind	repeating-record
Type	Repeating record with columns: String, Float, Float, Float
Default	Required / No default
Deprecated	No

Structure of the system is the system is not extended (i. e. is a molecule or cluster). The coordinates are in Angstroms, and have xyz format, i.e., Atomic_symbol1 x y z Atomic_symbol2 x y z . . .

Schema

|X| S F F F
|X| . . .

11.14 fermi_shift

Keyword	fermi_shift
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Shift of the fermi-energy in eV.

Schema

|X| F

11.15 ismetal

Keyword	ismetal
Kind	scalar
Type	Logical
Default	False
Deprecated	No

Toggle whether the system should be treated as a metal or as an insulator.

Schema

False |X| L

11.16 multiplicity

Keyword	multiplicity
Kind	scalar
Type	Integer
Default	0
Deprecated	No

Multiplicity of the system.

Schema

0 |X| I

11.17 number_of_atoms

Keyword	number_of_atoms
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Number of atoms in the system

Schema

|X| I

11.18 polarization

Keyword	polarization
Kind	repeating-record
Type	Repeating record with columns: Float, Float, Float
Default	Required / No default
Deprecated	No

Set any number of different polarization directions to calculate in cartesian coordinates. Example: set to x, y, z, i.e., polarization 1.0 0.0 0.0 0.0 1.0 0.0 0.0 0.0 1.0

If set to 0.0 0.0 0.0 (default), only polarization average will be calculated.

Schema

0.0 0.0 0.0 |X| F F F
|X| ...

11.19 species

Keyword	species
Kind	vector
Type	Vector: String, Integer
Default	Required / No default
Deprecated	No

chemical species and number of each species

Schema

|X| S I

11.20 spectral_broadening

Keyword	spectral_broadening
Kind	scalar
Type	Float
Default	0.0
Deprecated	No

Broadening used in some spectroscopic methods.

Schema

0.0 |X| F

11.21 spin_moment

Keyword	spin_moment
Kind	repeating-record
Type	Repeating record with columns: Integer, Float
Default	Required / No default
Deprecated	No

spin moments of atomic types set spin moment for ferromagnetic systems. Only allows one moment per chemical element. Z spin_moment

Schema

|X| I F
|X| ...

11.22 supercell_dimensions

Keyword	supercell_dimensions
Kind	vector
Type	Vector: Integer, Integer, Integer
Default	Required / No default
Deprecated	No

supercell dimensions (number of cells in each direction).

Schema

|X| I I I

11.23 vasp_outcar

Keyword	vasp_outcar
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

specify outcar file to use as input to corvus.

Schema

|X| S

11.24 vasp_snapshot

Keyword	vasp_snapshot
Kind	scalar
Type	Integer
Default	-1
Deprecated	No

specify index of structure (snapshot if MD run) to use.

Schema

-1 |X| I

11.25 vasp_xdatcar

Keyword	vasp_xdatcar
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

specify xdatcar file to use as input to corvus.

Schema

|X| S

11.26 vasp_xml

Keyword	vasp_xml
Kind	scalar
Type	String

Default	Required / No default
Deprecated	No

specify vasp output xml file to use as input to corvus.

Schema

|X| S

11.27 xyz_input

Keyword	xyz_input
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

xyz input file name

Schema

|X| S

11.28 xyz_snapshot

Keyword	xyz_snapshot
Kind	scalar
Type	Integer
Default	1
Deprecated	No

for multiple structure xyz files, use this snapshot.

Schema

1 |X| I

12 Abinit

12.1 abinit.ng2qpt

Keyword	abinit.ng2qpt
Kind	vector
Type	Vector: Integer, Integer, Integer
Default	Required / No default

Deprecated	No
------------	----

Equivalent to Abinit's (anaddb) ng2qpt input variable: Defines the number of q-point perturbations in each direction of the finer grid for density functional perturbation theory (DFPT) simulations. Please refer to the Abinit manual for further details.

Schema

|X| I I I

12.2 abinit.ngqpt

Keyword	abinit.ngqpt
Kind	vector
Type	Vector: Integer, Integer, Integer
Default	Required / No default
Deprecated	No

Equivalent to Abinit's ngqpt input variable: Defines the number of q-point perturbations in each direction of the grid for density functional perturbation theory (DFPT) simulations, and is that equivalent to the general "nqpoints" Corvus input variable. Please refer to the Abinit manual for further details.

NOTE: This code-specific input variable will be replaced by the general "nqpoints" variable in the future.

Schema

|X| I I I

12.3 abinit.verbatim

Keyword	abinit.verbatim
Kind	scalar
Type	Free-form
Default	Required / No default
Deprecated	No

The content of this input variable is passed as is (i. e. "verbatim") into the input of all Abinit calculations, and it is meant to help with any Abinit input that is currently not implemented in Corvus. Thus, it should be used carefully to avoid inconsistencies with the automatically generated input. Users should refer to the Abinit manual for information on this extra input.

Schema

|X| P

13 DMDW

13.1 dmdw.ioflag

Keyword	dmdw.ioflag
Kind	scalar
Type	Free-form
Default	Required / No default
Deprecated	No

Set the amount of output printed by DMDW. Possible values are: 0: Terse, prints out only the desired result (σ^2 , u^2 , etc). 1: Verbose, prints out the pole frequencies and weights, as well as estimates of the Einstein temperatures for each path. Please refer to the DMDW section of the Feff manual for further details.

NOTE: The current format described below is temporary for compatibility with previous versions of Corvus. This will be changed to "Integer" in the future.

Schema

|X| P

13.2 dmdw.nlanc

Keyword	dmdw.nlanc
Kind	scalar
Type	Free-form
Default	Required / No default
Deprecated	No

Set the number of Lanczos poles (i. e. iterations) in DMDW. Larger values usually improve convergence of the target quantity. However, the number of poles should not exceed the dimensions of the subspace spanned by the projection of the path into the appropriate eigenmodes of the Hessian. This means that this variable should always be less than $3*N-6$, where N is the number of atoms in the system. In practice, a value of 6-8 is sufficient to obtain converged mean square relative displacements (MSRD, or σ^2) for EXAFS. For crystallographic mean square displacements (MSD, or u^2) at least 16 poles are usually needed. Please refer to the DMDW section of the Feff manual for further details.

NOTE: The current format described below is temporary for compatibility with previous versions of Corvus. This will be changed to "Integer" in the future.

Schema

|X| P

13.3 dmdw.paths

Keyword	dmdw.paths
Kind	scalar
Type	Free-form
Default	Required / No default
Deprecated	No

Set up the path descriptors that will generate the list of paths for which properties will be calculated. The list has the following form: <Number of descriptors> <Descriptor 1> <Descriptor 2> . .

Each descriptor has the form: <Number of atoms in path> <Atom index 1> ... <Max. path length (Ang)>

The atom indices can take the value 0 which acts as a wildcard for all atoms in the system.

Examples:

2 2 1 0 3.0 3 2 0 5 6.0

This section defines 2 paths descriptors. The first one generates all paths with two atoms, starting in atom 1 and going to all other atoms in the systems, but subject to a maximum effective path length of 3.0 Ang. The second one generates all paths with three atoms, starting in atom 2 and ending in atom 5, while passing through all other atoms in the system, but with a maximum effective length of 6.0 Ang. Using the same syntax atoms can be selected to compute their u^2 . For example the paths section

1 1 0 0.0

will produce u^2 for all atoms in the system. Please refer to the DMDW section of the Feff manual for further details.

NOTE: The current format described below is temporary for compatibility with previous versions of Corvus.

Schema

|X| P

13.4 dmdw.tempgrid

Keyword	dmdw.tempgrid
Kind	scalar
Type	Free-form
Default	Required / No default
Deprecated	No

Set up the temperature grid to compute the thermal properties in DMDW. It has the following form:

<Number of temperature> <Min. temp.> <Max. temp.>

If the number of desired temperatures is just one, the maximum temperature input is not needed. Please refer to the DMDW section of the Feff manual for further details.

NOTE: The current format described below is temporary for compatibility with previous versions of Corvus.

Schema

|X| P

14 NWChem

14.1 nwchem.basis

Keyword	nwchem.basis
Kind	repeating-record
Type	Repeating record with columns: String
Default	Required / No default
Deprecated	No

Set the Gaussian basis set to be used in the NWChem calculations. The format is the same as in NWChem:

<Atom label> <Basis set name> . .

Please refer to the NWChem manual for further details.

Schema

|X| S ...

|X| ...

14.2 nwchem.charge

Keyword	nwchem.charge
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Set the net charge of the system for an NWChem DFT simulation. NOTE: This input variable will be superseded in the future by the more general "charge" variable.

Please refer to the NWChem manual for further details.

Schema

|X| S

14.3 nwchem.iter

Keyword	nwchem.iter
Kind	scalar
Type	String
Default	Required / No default

Deprecated	No
------------	----

Schema

|X| S

14.4 nwchem.mult

Keyword	nwchem.mult
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Set the multiplicity of the system for an NWChem DFT simulation. NOTE: This input variable will be superseded in the future by the more general "multiplicity" variable.

Please refer to the NWChem manual for further details.

Schema

|X| S

14.5 nwchem.qmd.dt_nucl

Keyword	nwchem.qmd.dt_nucl
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Set the time step for nuclear motion in a quantum MD simulation.

Please refer to the NWChem manual for further details.

Schema

|X| S

14.6 nwchem.qmd.nstep_nucl

Keyword	nwchem.qmd.nstep_nucl
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Set the number of nuclear motion steps in a quantum MD simulation.

Please refer to the NWChem manual for further details.

Schema

|X| S

14.7 nwchem.qmd.print_xyz

Keyword	nwchem.qmd.print_xyz
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Toggle printing of xyz coordinates in a quantum MD simulation.
Please refer to the NWChem manual for further details.

Schema

|X| S

14.8 nwchem.qmd.snapstep

Keyword	nwchem.qmd.snapstep
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

|X| S

14.9 nwchem.qmd.targ_temp

Keyword	nwchem.qmd.targ_temp
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Set the target temperature in a quantum MD simulation.
Please refer to the NWChem manual for further details.

Schema

|X| S

14.10 nwchem.qmd.thermostat

Keyword	nwchem.qmd.thermostat
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Set the thermostat type to be used in a quantum MD simulation.
Please refer to the NWChem manual for further details.

Schema

|X| S

14.11 nwchem.xas.alpha

Keyword	nwchem.xas.alpha
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

|X| S

14.12 nwchem.xas.nroots

Keyword	nwchem.xas.nroots
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

|X| S

14.13 nwchem.xas.vec

Keyword	nwchem.xas.vec
Kind	scalar
Type	String
Default	Required / No default

Deprecated	No
------------	----

Schema

|X| S

14.14 nwchem.xas.xrayenergywin

Keyword	nwchem.xas.xrayenergywin
Kind	vector
Type	Vector: Float, Float
Default	Required / No default
Deprecated	No

Schema

|X| F F

14.15 nwchem.xaselem

Keyword	nwchem.xaselem
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

|X| S

14.16 nwchem.xc

Keyword	nwchem.xc
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Set the exchange correlation potential for an NWChem DFT simulation. NOTE: This input variable will be superseded in the future by the more general "xc" variable.

Please refer to the NWChem manual for further details.

Schema

|X| S

15 FEFF

15.1 debyetemp

Keyword	debyetemp
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

debyetemp temp

Specify the Debye Model temperature (in K) for the calculation of EXAFS DW factors.

Schema

|X| F

15.2 dmdw_nlanczos

Keyword	dmdw_nlanczos
Kind	scalar
Type	Integer
Default	6
Deprecated	No

dmdw_nlanczos nLanczos_Interations

Specify the number of Lanczos iterations to be done for the calculation of EXAFS DW factors with the dynamical matrix method.

Schema

6 |X| I

15.3 feff.absolute

Keyword	feff.absolute
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

feff.absolute UseAbsoluteUnits Print spectrum in absolute units instead of normalizing.

Schema

|X| L

15.4 feff.afolp

Keyword	feff.afolp
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

feff.afolp folpx Set maximum overlap for automatic overlap search. folpx - maximum overlap allowed.

Schema

|X| F

15.5 feff.atoms

Keyword	feff.atoms
Kind	repeating-record
Type	Repeating record with columns: Float, Float, Float, Integer
Default	Required / No default
Deprecated	No

feff.atoms x1 y1 z1 ipot1 x2 y2 z2 ipot2 . . .

Specify atomic positions in cartesian coordinates (in Angstroms) and unique potential indices of each atom in the cluster, one atom per line.

Schema

|X| F F F I
|X| . . .

15.6 feff.cfaverage

Keyword	feff.cfaverage
Kind	vector
Type	Vector: Integer, Integer, Float
Default	Required / No default
Deprecated	No

Obsolete: Do not use.

Schema

|X| I I F

15.7 feff.cgrid

Keyword	feff.cgrid
Kind	vector
Type	Vector: Float, Integer, Integer, Integer, Integer
Default	Required / No default
Deprecated	No

Not implemented yet.

Schema

|X| F I I I I

15.8 feff.chbroad

Keyword	feff.chbroad
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

feff.chbroad ichbroad Set method of calculating core-hole lifetime broadening. ichbroad - 0: Calculate Green's function at energy with imaginary part equal to $\Gamma_{CH}/2$. 1: Convolve final spectrum with Lorentzian of full width at half-max Γ_{CH} .

Schema

|X| I

15.9 feff.chsh

Keyword	feff.chsh
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

feff.chsh ich Correct chemical shift. ich = 0 or 1.

Schema

|X| I

15.10 feff.chwidth

Keyword	feff.chwidth
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

feff.chwidth gamma_ch Set core-hole width instead of using FEFF's internal table of widths.

Schema

|X| F

15.11 feff.cif

Keyword	feff.cif
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Specify a cif input file. Dangerous. Use cif_input instead.

Schema

|X| S

15.12 feff.compton

Keyword	feff.compton
Kind	vector
Type	Vector: Float, Integer, Integer
Default	Required / No default
Deprecated	No

Not implemented yet.

Schema

|X| F I I

15.13 feff.config

Keyword	feff.config
Kind	repeating-record
Type	Repeating record with columns: String

Default	Required / No default
Deprecated	No

Not implemented yet.

Schema

|X| S
|X| ...

15.14 feff.control

Keyword	feff.control
Kind	vector
Type	Vector: Integer, Integer, Integer, Integer, Integer, Integer
Default	Required / No default
Deprecated	No

feff.control ipot ixsph ifms ipaths igenfmt iff2x

feff.control lets you run one or more of the feff program modules separately. There is a switch for each of six parts of feff: 0 means not to run that module, 1 means to run it.

Schema

|X| I I I I I I

15.15 feff.coordinates

Keyword	feff.coordinates
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

feff.coordinates i i must be an integer from 1 through 6. It specifies the units of the atoms of the unit cell given in the ATOMS card for reciprocal space calculations. If the card is omitted, the default value icoord = 3 is assumed. FIX check this 1. Cartesian coordinates, Angstrom units. Like feff - you can copy from a real-space feff.inp file if your lattice vectors coincide with atoms in that feff.inp file. 2. Cartesian coordinates, fractional units (i.e., fractions of the lattice vectors ; should be numbers between 0 and 1). Similar to feff. 3. Cartesian coordinates, units are fractional with respect to FIRST lattice vector. Like SPRKKR. (default) 4. Given in lattice coordinates, in fractional units. Like WIEN2k (but beware of some ‘funny’ lattice types, e.g. rhombohedral, in WIEN2k case.struct if you’re copy-pasting) 5. Given in lattice coordinates, units are fractional with respect to FIRST lattice vector. 6. Given in lattice coordinates, Angstrom units.

Schema

|X| I

15.16 feff.corehole

Keyword	feff.corehole
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

feff.corehole CoreHoleApproximation Set method for calculating corehole interaction. CoreHoleApproximation - None: Do not use a corehole. FSR : Final-state rule RPA : Use RPA dielectric function to calculate screening of core-hole.

Schema

|X| S

15.17 feff.corrections

Keyword	feff.corrections
Kind	vector
Type	Vector: Float, Float
Default	Required / No default
Deprecated	No

feff.corrections vrcorr vicorr Correct the Fermi cutoff and broadening in the final spectrum calculation. vrcorr - Shift Fermi cutoff by -vrcorr. vicorr - Add extra Lorentzian broadening with half width at half max of vicorr.

Schema

|X| F F

15.18 feff.corval

Keyword	feff.corval
Kind	vector
Type	Vector: Float, Float
Default	Required / No default
Deprecated	No

feff.corval emin emax The core-valence separation energy is found by scanning the DOS within an energy window. The emin parameter sets the lower bound (in eV) of this energy window and emax the upper. They are -70 eV and -20 eV by default. For some materials it is necessary to change these bounds, e.g. to -100 eV and -30 or -10 eV. For example, when SCF convergence is elusive because occupation numbers for one or more l-values are changing drastically between SCF iterations due to states moving above and below a poor estimate of the core-valence separation energy. We plan to replace the current mechanism by a more robust and automated algorithm, but in the meantime users can use the CORVAL card to handle some of these difficult cases.

Schema

|X| F F

15.19 feff.criteria

Keyword	feff.criteria
Kind	vector
Type	Vector: Float, Float
Default	Required / No default
Deprecated	No

feff.criteria critcw critpw Cutoff criteria for the path expansion filtering. critcw - tolerance for curved wave expansion critpw - tolerance for initial plane wave approximation

Schema

|X| F F

15.20 feff.crpa

Keyword	feff.crpa
Kind	vector
Type	Vector: Integer, Float
Default	Required / No default
Deprecated	No

Not currently working.

Schema

|X| I F

15.21 feff.danes

Keyword	feff.danes
Kind	vector
Type	Vector: Float, Float, Float
Default	Required / No default
Deprecated	No

feff.danes xkmax xkstep estep Calculate diffraction anomalous near edge structure. xkmax - calculate up to $k = xkmax$ xkstep - use steps of size xkstep estep - near the edge, use steps calculated from estep

Schema

|X| F F F

15.22 feff.debye

Keyword	feff.debye
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

feff.debye Temperature Debye-Temperature idwopt Set temperature and Debye temperature for calculations of EXAFS Debye-Waller factors. idwopt - set method for calculating DW factors. NOTE: This input variable is not fully implemented yet.

Schema

```
|X| S ...
```

15.23 feff.density

Keyword	feff.density
Kind	block
Type	Multi-line heterogeneous block
Default	Required / No default
Deprecated	No

Not implemented yet.

Schema

```
# den_type, grid_type, outfile
|X| S S S
# origin
|X| F F F
# number of points in each direction
|X| I I I
# vectors along which to step
|X| F F F
|X| F F F
|X| F F F
```

15.24 feff.dims

Keyword	feff.dims
Kind	vector
Type	Vector: Integer, Integer
Default	Required / No default
Deprecated	No

feff.dims nmax lmax Set maximum dimensions for fms calculations. nmax - maximum number of atoms in fms matrix. lmax - maximum angular momentum of fms matrix.

Schema

|X| I I

15.25 feff.dmdw

Keyword	feff.dmdw
Kind	vector
Type	Vector: String, Integer, Integer, Integer
Default	Required / No default
Deprecated	No

NOTE: This input variable is not fully implemented yet.

Schema

|X| S I I I

15.26 feff.edge

Keyword	feff.edge
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

feff.edge edgeLabel Set edge to calculate for XAS, XES, EELS, and other related spectra.

Schema

|X| S ...

15.27 feff.egapse

Keyword	feff.egapse
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

feff.egapse egap Use this gap energy when applying the many-pole self-energy.

Schema

|X| F

15.28 feff.egrid

Keyword	feff.egrid
Kind	repeating-record
Type	Repeating record with columns: String, String, Float, Float
Default	Required / No default
Deprecated	No

This card can be used to customize the energy grid. The EGRID card is followed by lines specifying the type of grid, minimum and maximum values for the grid, and the grid step, i.e.

grid_type grid_min grid_max grid_step

The grid type parameter is a string that can take the values e_grid, k_grid, or exp_grid. When using the e_grid or k_grid grid types, grid_min, grid_max, and grid_step are given in eV or $^{-1}$ respectively. For the exp_grid type, grid_min and grid_max are the minimum and maximum grid values in eV, and grid_step is the exponential, i.e.

$$E_i = \text{grid_min} + (\text{grid_max} - \text{grid_min})[\exp(\text{grid_step} \times i) - 1].$$

A fourth grid type user grid is also available for feff. user grid is followed by an arbitrary number of lines, each specifying an energy point in eV, i.e., user_grid 0.1 1.5 3.45 6.0 . . .

Schema

```
|X| S S F F
|X| ...
```

15.29 feff.ellipticity

Keyword	feff.ellipticity
Kind	vector
Type	Vector: Float, Float, Float, Float
Default	Required / No default
Deprecated	No

feff.ellipticity elpty x y z Set the ellipticity of the x-rays. This card is used with the feff.polarization. elpty - ratio of amplitudes of electric field in the two orthogonal directions of elliptically polarized light. Only the absolute value of the ratio is important for nonmagnetic materials. The present code can distinguish left- and right-circular polarization only with the feff.xmcd or feff.xncd. A zero value of the ellipticity corresponds to linear polarization, and unity to circular polarization. The default value is zero. x, y, z - coordinates of any nonzero vector in the direction of the incident beam. This vector should be approximately normal to the polarization vector

Schema

```
|X| F F F F
```

15.30 feff.elnes

Keyword	feff.elnes
Kind	repeating-record
Type	Repeating record with columns: Float, Float, Float
Default	Required / No default
Deprecated	No

This card is not implements yet.

Schema

```
|X| F F F
|X| S ...
|X| ...
# In future, fix so that we actually have these params, not just strings.
```

15.31 feff.eps0

Keyword	feff.eps0
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

feff.eps0 eps0 Set dielectric constant used for many-pole self-energy calculations.

Schema

```
|X| F
```

15.32 feff.equivalence

Keyword	feff.equivalence
Kind	scalar
Type	Integer
Default	0
Deprecated	No

feff.equivalence ieq This optional card is only active in combination with the CIF card. It tells feff how to generate potential types from the list of atom positions in the ‘cif’ file. If ieq = 1, the crystallographic equivalence as expressed in the ‘cif’ file is respected; that is, every separate line containing a generating atom position will lead to a separate potential type. This means that, e.g., in HOPG graphite, the two generating positions will give rise to two independent C potentials. This is also the default behavior if the EQUIVALENCE card is not specified. If ieq = 2, unique potentials are assigned based on atomic number Z only. That is, all C atoms will share a C potential and so on. This is how most feff calculations are run. Whether it is sensible or not to do this depends on

the system and on the property one wishes to calculate. Keep in mind that feff is a muffin tin code, and may therefore be indifferent to certain differences between crystallographically inequivalent sites. On the other hand, if an element occurs in the crystal with different oxidation states, it may be necessary to assign separate potentials to these different types in order to describe the crystal properly and get accurate spectra. If $ieq = 3$, unique potentials are assigned based on atomic number Z and the first shell. This can be useful e.g. to treat larger systems with crystal defects, where only first neighbors of the defect need to be treated differently from all more distant atoms of a certain Z . (To be implemented.) If $ieq = 4$, a hybrid of methods 1 and 2 is used. That is, if the number of unique crystallo- graphic positions does not exceed a hard-coded limit ($nphx=9$ in the current version), they are treated with the correct crystallographic equivalence. If the number of unique crystallographi- cally inequivalent sites is larger, they get combined by atomic number Z . This ad hoc approach is a practical way of simply limiting the number of unique potentials. This makes sense be- cause, first of all, there are certain hardcoded limits that would require recompilation of the code, requiring more RAM memory and more work than a user may want to do.

Schema

0 |X| I

15.33 feff.equivalence_nmax

Keyword	feff.equivalence_nmax
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Schema

|X| I

15.34 feff.exafs

Keyword	feff.exafs
Kind	vector
Type	Vector: Float, Float
Default	Required / No default
Deprecated	No

feff.exafs $xkmax$ $xkstep$ Calculate EXAFS spectrum. $xkmax$ - calculate spectrum up to $k = xkmax$ $xkstep$ - calculate in steps of $xkstep$

Schema

|X| F F

15.35 feff.exchange

Keyword	feff.exchange
Kind	vector
Type	Vector: Integer, Float, Float, Integer
Default	Required / No default
Deprecated	No

feff.exchange iexc vr vi iexc0 Use feff.exchange to change the self-energy used in x-ray absorption calculations. ixc is an index specifying the potential model to use for the fine structure, and the optional ixc0 is the index of the model to use for the background function. The calculated potential can be corrected by adding a constant shift to the Fermi level given by vr0 and to a pure imaginary "optical" potential (i.e., uniform decay) given by vi0. Typical errors in Feff's self-consistent Fermi level estimate are about 1 eV. (The feff.corrections input is similar but allows the user to make small changes in vi0 and vr0 after the rest of the calculation is completed, for example in a fitting process.)

Indices for the available exchange models: 0 Hedin-Lundqvist + a constant imaginary part 1 Dirac-Hara + a constant imaginary part 2 ground state + a constant imaginary part 3 Dirac-Hara + HL imag part + a constant imaginary part 5 Partially nonlocal: Dirac-Fock for core + HL for valence electrons + a constant imaginary part

Schema

|X| I F F I

15.36 feff.exelfs

Keyword	feff.exelfs
Kind	repeating-record
Type	Repeating record with columns: Float
Default	Required / No default
Deprecated	No

This card is not implemented yet.

Schema

|X| F

|X| S

|X| ...

In future, fix so that we actually have these params, not just strings.

15.37 feff.extpot

Keyword	feff.extpot
Kind	scalar

Type	Logical
Default	Required / No default
Deprecated	No

Not currently working. Do not use.

Schema

|X| L

15.38 feff.fms

Keyword	feff.fms
Kind	vector
Type	Vector: Float, Integer, Integer, Float, Float, Float
Default	Required / No default
Deprecated	No

feff.fms rfms lfms2 minv toler1 toler2 rdirec Use full multiple-scattering. rfms - radius of cluster to use for fms calculation. lfms1 - 0 for solids, 1 for molecules minv - set algorithm for matrix inversion 0: LU decomposition 2: Lanczos 3: Broyden (less reliable) toler1 - tolerance to stop recursion and Broyden algorithm toler2 - sets the matrix element of the Gt matrix to zero if its value is less than toler2 rdirec - sets the matrix element of the Gt matrix to zero if the distance between atoms is larger than rdirec

Schema

|X| F I I F F F

15.39 feff.folp

Keyword	feff.folp
Kind	repeating-record
Type	Repeating record with columns: Integer, Float
Default	Required / No default
Deprecated	No

feff.folp ipot overlap Set the overlap for unique potential ipot.

Schema

|X| I F
|X| ...

15.40 feff.fprime

Keyword	feff.fprime
Kind	vector
Type	Vector: Float, Float, Float
Default	Required / No default
Deprecated	No

feff.fprime emin emax estep emin - minimum energy of calculation. emax - maximum energy of calculation estep - energy step of calculation Calculate atomic scattering factor f'

Schema

|X| F F F

15.41 feff.hole

Keyword	feff.hole
Kind	vector
Type	Vector: Integer, Float
Default	Required / No default
Deprecated	Yes

DEPRECATED: Use feff.edge instead.

Specify the edge using the hole number ihole, e.g., K-edge : ihole = 1 L1-edge: ihole = 2 s02 specifies the EXAFS amplitude reduction factor, and should be set to 1.

Schema

|X| I F

15.42 feff.hubbard

Keyword	feff.hubbard
Kind	vector
Type	Vector: Float, Float, Float, Integer
Default	Required / No default
Deprecated	No

Not currently working.

Schema

|X| F F F I

15.43 feff.interstitial

Keyword	feff.interstitial
Kind	vector
Type	Vector: Integer, Float, Float
Default	Required / No default
Deprecated	No

feff.interstitial inters vtot vint_scl The construction of the interstitial potential and density may be changed by using this card. inters = 1 might be useful when only the surroundings of the absorbing atom are specified in 'feff.inp'. inters defines how to find the interstitial potential. inters=0: The interstitial potential is found by averaging over the entire extended cluster in 'feff.inp'. (Default) inters=1: the interstitial potential is found locally around the absorbing atom. vtot: the volume per atom normalized by ratmin3 ($vtot = (\text{volume per atom}) / \text{ratmin3}$), where ratmin is the shortest bond for the absorbing atom. This quantity defines the total volume (needed to calculate the interstitial density) of the extended cluster specified in 'feff.inp'. If $vtot \leq 0$ then the total volume is calculated as a sum of Norman sphere volumes. Otherwise, total volume = $nat * (vtot * \text{ratmin3})$, where nat is the number of atoms in an extended cluster.

Schema

|X| I F F

15.44 feff.ion

Keyword	feff.ion
Kind	repeating-record
Type	Repeating record with columns: Integer, Float
Default	Required / No default
Deprecated	No

feff.ion ipot ionization feff.ion ionizes all atoms with unique potential index ipot. NOTE: This input variable is not fully implemented yet.

Schema

|X| I F
|X| ...

15.45 feff.iorder

Keyword	feff.iorder
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

feff.iorder iorder Set order of approximation when calculating effective scattering matrices.

Schema

|X| I

15.46 feff.jumprm

Keyword	feff.jumprm
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

feff.jumprm UseRemoveJump If true, smooth the jump between muffin tin potentials and interstitial.

Schema

|X| L

15.47 feff.kmesh

Keyword	feff.kmesh
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Specify the kmesh. feff.kmesh nkp(x) [nkpy nkpz [ktype [usesym]]] This card specifies the mesh of k-vectors used to sample the full Brillouin Zone for the evaluation of Brillouin Zone integrals. Nkp is the number of points used in the full zone. It can be specified either as "nkpx nkpy nkpz", "nkp", or "nkp 0 0". If usesym = 1, the zone is reduced to its irreducible wedge using the symmetry options specified in file symfile, which must be present in the working directory. The k-mesh is constructed using the tetrahedron method of Bloechl et al., Phys. Rev. B, 1990. The parameter ktype is meant for time-saving only and means: ktype=1 : regular mesh of nkp points for all modules ktype=2 : use nkp points for ldos/fms and nkp/5 points for pot (significant time savings) ktype=3 : use nkp points for ldos/fms and nkp/5 points for pot (near edge) ; reduce nkp for all modules as we get away from near-edge (somewhat experimental) * use a k-mesh of 1000 points in the full BZ. feff.kmesh 1000 * use a k-mesh of 10x5x3 points for a large, irregular cell. feff.kmesh 10 5 3 * use a k-mesh of 1000 points and try to save time: feff.kmesh 1000 0 0 3

Schema

|X| I ...

15.48 feff.lattice

Keyword	feff.lattice
Kind	repeating-record
Type	Repeating record with columns: String
Default	Required / No default
Deprecated	No

This card specifies the lattice. First, its type must be specified using a single letter : P for primitive, F for face centered cubic, I for body centered cubic, H for hexagonal. The following three lines give the three basis vectors in Carthesian Angstrom coordinates. They are multiplied by scale (e.g., 0.529177 to convert from bohr to Angstrom). feff.lattice P 3.18800 ax ay az bx by bz cx cy cz

Schema

```
# Need to fix this so that all parameters are set in separate fields.
|X| S
|X| ...
```

15.49 feff.ldec

Keyword	feff.ldec
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

For use in calculations of NRIXS. Not implemented yet.

Schema

```
|X| I
```

15.50 feff.ldos

Keyword	feff.ldos
Kind	vector
Type	Vector: Float, Float, Float, Integer
Default	Required / No default
Deprecated	No

feff.ldos emin emax broadening npoints Run angular momentum project density of states calculations. emin - minimum energy of energy grid. emax - maximum energy of energy grid. broadening - Lorentzian broadening to use in calculation. npoints - number of energy points in grid.

Schema

```
|X| F F F I
```

15.51 feff.lfms1

Keyword	feff.lfms1
Kind	scalar
Type	Integer
Default	-1
Deprecated	No

set maximum angular momentum to use in potentials

Schema

-1 |X| I

15.52 feff.lfms2

Keyword	feff.lfms2
Kind	scalar
Type	Integer
Default	-1
Deprecated	No

set maximum angular momentum to use in FMS

Schema

-1 |X| I

15.53 feff.ljmax

Keyword	feff.ljmax
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

For use in calculations of NRIXS. Not implemented yet.

Schema

|X| I

15.54 feff.magic

Keyword	feff.magic
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

This card is not implemented yet.

Schema

|X| F

15.55 feff.mbconv

Keyword	feff.mbconv
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	Yes

Deprecated: Should not be used.

Schema

|X| L

15.56 feff.mpse

Keyword	feff.mpse
Kind	vector
Type	Vector: Integer, Integer
Default	Required / No default
Deprecated	No

feff.mpse ipl npl Use many-pole model self-energy. This requires the loss function, which can be provided externally, or approximated using feff.opcons. ipl - Set method for self-energy inclusion: 1: use an "average" self-energy which is applied to the whole system (default). 2: use a density dependent self-energy which is different at every point inside the muffin-tin radius.

Schema

|X| I I

15.57 feff.multipole

Keyword	feff.multipole
Kind	vector
Type	Vector: Integer, Integer
Default	Required / No default
Deprecated	No

feff.multipole le2 l2lp Set multipole expansion approximation. Specifies which multipole transitions to include in the calculations. The options are: only dipole (le2 = 0, default), dipole and magnetic dipole (le2 = 1), dipole and quadrupole (le2 = 2). This cannot be used with NRIXS and is not supported with EXELFS and ELNES. The additional field l2lp can be used to calculate individual dipolar contributions coming from $L \rightarrow L + 1$ (l2lp = 1) and from $L \rightarrow L - 1$ (l2lp = -1). Notice that in polarization dependent data there is also a cross term, which is calculated only when l2lp = 0

Schema

|X| I I

15.58 feff.nleg

Keyword	feff.nleg
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

feff.nleg nleg Set maximum number of legs to use in path expansion. A Single scattering path has 2 legs.

Schema

|X| I

15.59 feff.nohole

Keyword	feff.nohole
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	Yes

feffnohole UseNoHole Do not use a hole in the calculation. DEPRECATED - Use feff.corehole instead.

Schema

|X| L

15.60 feff.nrixs

Keyword	feff.nrixs
Kind	repeating-record
Type	Repeating record with columns: Integer
Default	Required / No default
Deprecated	No

Schema

|X| I

|X| F ...

|X| ...

15.61 feff.nstar

Keyword	feff.nstar
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

feff.nstar WriteNStar Write nstar.dat with effective coordination number N*.

Schema

|X| L

15.62 feff.numdens

Keyword	feff.numdens
Kind	repeating-record
Type	Repeating record with columns: Integer, Float
Default	Required / No default
Deprecated	No

Schema

|X| I F

|X| ...

15.63 feff.opcons

Keyword	feff.opcons
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

Calculate the dielectric function of the material within the atomic approximation. Very fast. Used to calculate loss.dat for use with feff.mpse self-energy calculations.

Schema

|X| L

15.64 feff.overlap

Keyword	feff.overlap
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

feff.overlap can be used to construct approximate overlapped atom potentials when atomic coordinates are not known or specified. NOTE: This input variable is not fully implemented yet.

Schema

|X| I

15.65 feff.pcriteria

Keyword	feff.pcriteria
Kind	vector
Type	Vector: Float, Float
Default	Required / No default
Deprecated	No

feff.pcriteria pcrith Set criteria for filtering in pathfinder. The keep-criterion pcrith looks at the amplitude of chi (in the plane wave approximation) for the current path and compares it to a single scattering path of the same effective length. To set this value, consider the maximum degeneracy you expect and divide your plane wave criterion by this number. For example, in fcc Cu, typical degeneracies are 196 for paths with large r, and the minimum degeneracy is 6. So a keep criterion of 0.08heap-criterion pcrith filters paths as the pathfinder puts all paths into a heap (a partially ordered data structure), then removes them in order of increasing total path length. Each path that is removed from the heap is modified and then considered again as part of the search algorithm. The heap filter is used to decide if a path has enough amplitude in it to be worth

further consideration. If a path can be eliminated at this point, entire trees of derivative paths can be neglected, leading to enormous time savings. This test does not come into play until paths with at least 4 legs are being considered, so single scattering and triangular (2 and 3 legged) paths will always pass this test. Because only a small part of a path is used for this criterion, it is difficult to predict what appropriate values will be.

Schema

|X| F F

15.66 feff.pmbse

Keyword	feff.pmbse
Kind	vector
Type	Vector: Integer, Integer, Integer, Integer
Default	Required / No default
Deprecated	No

Schema

|X| I I I I

15.67 feff.polarization

Keyword	feff.polarization
Kind	vector
Type	Vector: Float, Float, Float
Default	Required / No default
Deprecated	No

feff.polarization x y z Set polarization vector of x-rays in cartesian coordinates of input file.

Schema

|X| F F F

15.68 feff.potentials

Keyword	feff.potentials
Kind	repeating-record
Type	Repeating record with columns: Integer, Integer, String, Integer, Integer, Float
Default	Required / No default
Deprecated	No

feff.potentials ipot iz pot_label lfms1 lfms2 stoichiometry ... Set unique potentials for FEFF calculation.

Schema

```
|X| I I S I I F
|X| ...
```

15.69 feff.potentials.spin

Keyword	feff.potentials.spin
Kind	repeating-record
Type	Repeating record with columns: Integer, Float
Default	Required / No default
Deprecated	No

feff.potentials.spin ipot spin_moment ... Set spin moments of atoms defined in potentials card

Schema

```
|X| I F
|X| ...
```

15.70 feff.preps

Keyword	feff.preps
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

feff.preps PrintEpsilon Print dielectric function as calculated by feff.opcons.

Schema

```
|X| L
```

15.71 feff.print

Keyword	feff.print
Kind	vector
Type	Vector: Integer, Integer, Integer, Integer, Integer, Integer
Default	Required / No default
Deprecated	No

feff.pring ipr_pot ipr_xsph ipr_fms ipr_path ipr_genfmt ipr_ff2x Set print level for each module of FEFF: pot: 0 write 'pot.bin' only 1 add 'misc.dat' 2 add 'potNN.dat' 3 add 'atomNN.dat' xsph: 0 write 'phase.bin' and 'xsect.bin' only 1 add 'axafs.dat' and 'phase.dat' 2 add 'phaseNN.dat' and 'phminNN.dat' 3 add 'ratio.dat' (for XMCD normalization) and 'emesh.dat'. fms: 0 write 'gg.bin' 1 write 'gg.dat'

path: 0 write 'paths.dat' only 1 add 'crit.dat' 3 add 'fbeta' files (plane wave $|f(\beta)|$ approximations) 5 Write only 'crit.dat' and do not write 'paths.dat'. (This is useful genfmt 0 write 'list.dat', and write 'feff.bin' with all paths with importance greater than or equal to two thirds of the curved wave importance criterion

genfmt: 0 write 'list.dat', and write 'feff.bin' with all paths with importance greater than or equal to two thirds of the curved wave importance criterion 1 write all paths to 'feff.bin'

ff2x: 0 write 'chi.dat' and 'xmu.dat' 2 add 'chipNNNN.dat' (chi(k) for each path individually) 3 add 'feffNNNN.dat' and 'files.dat', and do not add 'chipNNNN.dat' files

Schema

|X| I I I I I I

15.72 feff.rconv

Keyword	feff.rconv
Kind	vector
Type	Vector: Float, String
Default	Required / No default
Deprecated	No

Advanced: Print running convolution with the spectral function.

Schema

|X| F S

15.73 feff.real

Keyword	feff.real
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

Advanced: Use real phase shifts.

Schema

|X| L

15.74 feff.reciprocal

Keyword	feff.reciprocal
Kind	scalar
Type	Logical

Default	Required / No default
Deprecated	No

Work in reciprocal space (k-space calculation).

Schema

|X| L

15.75 feff.restart

Keyword	feff.restart
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

Not implemented yet.

Schema

|X| L

15.76 feff.rgrid

Keyword	feff.rgrid
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Schema

|X| F

15.77 feff.rhozzp

Keyword	feff.rhozzp
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

Not implemented yet.

Schema

|X| L

15.78 feff.rixs

Keyword	feff.rixs
Kind	vector
Type	Vector: Float, Float, Float
Default	Required / No default
Deprecated	No

feff.RIXS gam_Ei gam_El xmu The RIXS card sets the parameters for the RIXS calculation. It must be used with a workflow or script to produce RIXS. gam_Ei Broadening to use in the incident energy direction.

gam_El Broadening to use in the energy loss direction.

xmu Fermi energy to use for the RIXS calculation, if not present, the fermi level calculated in the SCF step will be used.

Schema

|X| F F F

15.79 feff.rlprint

Keyword	feff.rlprint
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

Advanced used in RIXS calculations, but automated by corvus. Prints real scattering wavefunction R_l .

Schema

|X| L

15.80 feff.rmultiplier

Keyword	feff.rmultiplier
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

feff.rmultiplier rmult Multiply coordinates of all atoms by rmult, expanding or contracting the system.

Schema

|X| F

15.81 feff.rpath

Keyword	feff.rpath
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

feff.rpath rmax Set maximum path length for path expansion calculations of EXAFS, EXELFS, DAFS, etc.

Schema

|X| F

15.82 feff.rphases

Keyword	feff.rphases
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

feff.rphases UseRealPhases Set real phase shift approximation. UseRealPhases - use real phase shift if true.

Schema

|X| L

15.83 feff.rsigma

Keyword	feff.rsigma
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

feff.rsigma UseRealSelfEnergy If true use only real part of self-energy.

Schema

|X| L

15.84 feff.s02

Keyword	feff.s02
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

feff.s02 s02 Set EXAFS amplitude reduction factor.

Schema

|X| F

15.85 feff.scf

Keyword	feff.scf
Kind	vector
Type	Vector: Float, Integer, Integer, Float, Integer
Default	Required / No default
Deprecated	No

feff.scf rfms1 lfms1 nscmt ca nmix Use self-consistent field calculation of potentials. rfms1 - radius of cluster to use for scf calculations. lfms1 - 0 for solids, 1 for molecules nscmt - maximum number of iterations in SCF calculation. ca - convergence factor for Broyden algorithm nmix - number of initial iterations of mixing algorithm to use.

Schema

|X| F I I F I

15.86 feff.scframp

Keyword	feff.scframp
Kind	vector
Type	Vector: Float, Integer
Default	Required / No default
Deprecated	No

Ramp scf radius up to final radius, starting from start_radius, and taking n_ramp steps.
SCFRAMP start_radius n_ramp

Schema

|X| F I

15.87 feff.screen

Keyword	feff.screen
Kind	repeating-record
Type	Repeating record with columns: String, String
Default	Required / No default
Deprecated	No

The screen module, which calculates the RPA core hole potential, is a 'silent' module: it has no obvious input but instead runs entirely on default values. Using the feff.SCREEN card you can change these default values. They will be written to an optional 'screen.inp' file (which you can also edit manually). The feff.SCREEN card can occur more than once in 'feff.inp'; all entries will be applied to the calculation. parameter must be one of : ner (40), nei (20), maxl (4), irrh (1), iend (0), lfx (0), emin (-40 eV), emax (0 eV), eimax (2 eV), ermin (0.001 eV), rfms (4.0), nrptx0 (251). For most calculations the default values given (between brackets) are fine. Occasionally we've changed rfms, maxl, or emin. Note that the screen is only active with feff.COREHOLE RPA . * Set the cluster radius for the RPA potential calculation higher than the default of 4.0 feff.COREHOLE RPA feff.SCREEN rfms 5.0

Schema

```
# We will also want to improve this interface.
|X| S S
|X| ...
```

15.88 feff.scxc

Keyword	feff.scxc
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Schema

```
|X| I
```

15.89 feff.self

Keyword	feff.self
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

feff.self PrintSelfEnergy Print out quasiparticle self-energy calculated during spectral-function convolution calculations.

Schema

|X| L

15.90 feff.setedge

Keyword	feff.setedge
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

feff.setedge UseEdgeTable Use table of experimental edge energies.

Schema

|X| L

15.91 feff.sfconv

Keyword	feff.sfconv
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

feff.sfconv UseSpectralFunctionConvolution If true, convolve spectrum with spectral function to account for multi-electron excitations.

Schema

|X| L

15.92 feff.sfse

Keyword	feff.sfse
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

feff.sfse k Print out off shell self-energy $\Sigma(k,E)$ calculated during spectral-function convolution calculations.

Schema

|X| F

15.93 feff.sgroup

Keyword	feff.sgroup
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

Specify space group by number (1 - 230).

Schema

|X| L

15.94 feff.sig2

Keyword	feff.sig2
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

feff.sig2 sig2 Set a single Debye-Waller factor for all paths, $\exp(-2 \cdot \text{sig2} \cdot k^2)$.

Schema

|X| F

15.95 feff.sig3

Keyword	feff.sig3
Kind	vector
Type	Vector: Float, Float
Default	Required / No default
Deprecated	No

feff.sig3 alphas thetas Set first and third cumulants for single scattering paths: alphas - first cumulant thetas - third cumulant

Schema

|X| F F

15.96 feff.siggk

Keyword	feff.siggk
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Not implemented yet.

Schema

|X| F

15.97 feff.spin

Keyword	feff.spin
Kind	vector
Type	Vector: Integer, Float, Float, Float
Default	Required / No default
Deprecated	No

feff.spin ispin sx sy sz Specify the type of spin-dependent calculation for spin along the (x, y, z) direction, along the z-axis by default. The SPIN card is required for the calculation of all spin-dependent effects, including XMCD and SPXAS. ispin: 2 spin-up SPXAS and LDOS -2 spin-down SPXAS and LDOS 1 spin-up portion of XMCD calculations -1 spin-down portion of XMCD calculations

Schema

|X| I F F F

15.98 feff.ss

Keyword	feff.ss
Kind	repeating-record
Type	Repeating record with columns: Integer, Integer, Float, Float
Default	Required / No default
Deprecated	No

Set an

Schema

|X| I I F F
|X| ...

15.99 feff.strfac

Keyword	feff.strfac
Kind	vector
Type	Vector: Float, Float, Float
Default	Required / No default
Deprecated	No

feff.strfac eta gmax rmax This card gives three non-physical internal parameters for the calculation of the KKR structure factors : the Ewald parameter and a multiplicative cutoff factor for sums over reciprocal (gmax) and real space (rmax) sums. Multiplicative means the code makes a 'smart' guess of a cutoff radius, but if one suspects something fishy is going on, one can here e.g. use gmax=2 to multiply this guess by 2. Eta is an absolute number. Given the stability of the Ewald algorithm, it shouldn't be necessary to use this card. Its use is not recommended. Only active in combination with the feff.reciprocal card.

Schema

|X| F F F

15.100 feff.symmetry

Keyword	feff.symmetry
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

feff.symmetry UseSymmetry If false, turn off symmetry considerations in path expansion.

Schema

|X| L

15.101 feff.target

Keyword	feff.target
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

feff.target ic Specifies the location of the absorber atom for reciprocal space calculations. It is entry ic of the feff.ATOMS card if an feff.ATOMS card and feff.LATTICE card are used. In conjunction with the cif_input card it is entry ic the list of atoms as given in the '.cif' file (i.e., a list of the crystallographically inequivalent atom positions in the unit cell). The target needs to be specified also for NOHOLE calculations. Note that this cannot be specified in the feff.POTENTIALS list because periodic boundary conditions would then produce an infinite number of core holes. * calculate a spectrum for the second atom in the feff.ATOMS list or CIF file. feff.TARGET 2

Schema

|X| I

15.102 feff.tdlda

Keyword	feff.tdlda
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

feff.tdlda ifxc Use TDLDA to calculate x-ray absorption spectrum. ifxc - set algorithm for tdlda approximation 0: use static approximation for screening of the x-ray field 1: use approximate dynamic screening of x-ray field and core-hole.

Schema

|X| I

15.103 feff.temperature

Keyword	feff.temperature
Kind	vector
Type	Vector: Float, Integer
Default	Required / No default
Deprecated	No

feff.temperature etemp isfcxc The TEMP card sets the electronic temperature and the exchange-correlation potential. etemp = 0 (default). etemp is in eV. There are 4 different options for the exchange-correlation potential. isfcxc No temperature dependent exchange correlations 11 von-Barth Hedin 1971 (default) 12 Perdew-Zunger Explicitly temperature dependent exchange correlations 21 Perrot Dharma-Wardana 1984 22 KSDT (recommended)

Schema

|X| F I

15.104 feff.title

Keyword	feff.title
Kind	repeating-record
Type	Repeating record with columns: String
Default	Required / No default
Deprecated	No

Set title for this feff calculation.

Schema

|X| S ...
|X| ...

15.105 feff.tolscf

Keyword	feff.tolscf
Kind	vector
Type	Vector: Float, Float, Float
Default	Required / No default
Deprecated	No

set tolerances for scf convergence feff.tolscf fermi_tol charge_tol partial_charge_tol

Schema

|X| F F F

15.106 feff.unfreezef

Keyword	feff.unfreezef
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

feff.unreezef UnfreezeFOrbitals If true, allow f-orbitals to relax during SCF calculation.

Schema

|X| L

15.107 feff.xanes

Keyword	feff.xanes
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

feff.xanes xkmax Calculate XANES spectrum. xkmax - calculate up to k = xkmax

Schema

|X| F

15.108 feff.xes

Keyword	feff.xes
Kind	vector
Type	Vector: Float, Float, Float
Default	Required / No default
Deprecated	No

feff.xes emin emax estep Calculate x-ray emission spectrum. emin - minimum energy of calculation. emax - maximum energy of calculation estep - energy step of calculation

Schema

|X| F F F

15.109 feff.xmcd

Keyword	feff.xmcd
Kind	vector
Type	Vector: Float, Float, Float
Default	Required / No default
Deprecated	No

feff.xmcd xkmax xkstep estep Caluculate x-ray magnetic (or natural) circular dichroism xkmax - calculate up to k = xkmax xkstep - use steps of size xkstep estep - near the edge, use steps calculated from estep

Schema

|X| F F F

15.110 nuctemp

Keyword	nuctemp
Kind	scalar
Type	Float
Default	300.0
Deprecated	No

nuctemp temp

Specify the temperature (in K) for the nuclear motion (DW factors).

Schema

300.0 |X| F

15.111 opcons.drude

Keyword	opcons.drude
Kind	vector
Type	Vector: Float, Float
Default	Required / No default
Deprecated	No

Calculate a drude approximation for the low energy part of the spectrum in opcons.

Schema

|X| F F

15.112 opcons.usesaved

Keyword	opcons.usesaved
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

feff.opcons Calculate loss function using an atomic approximation.

Schema

|X| L

15.113 run_dmdw

Keyword	run_dmdw
Kind	scalar
Type	Logical
Default	False
Deprecated	No

run standalone dmdw along with feff runs.

Schema

False |X| L

16 ORCA

16.1 orca.basis.basis

Keyword	orca.basis.basis
Kind	scalar
Type	String
Default	def2-TZVP
Deprecated	No

Schema

def2-TZVP |X| S

16.2 orca.basis.decontract

Keyword	orca.basis.decontract
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

Schema

|X| L

16.3 orca.ci

Keyword	orca.ci
Kind	scalar
Type	Logical
Default	F
Deprecated	No

Schema

F |X| L

16.4 orca.citype

Keyword	orca.citype
Kind	scalar
Type	String
Default	CCSD
Deprecated	No

Schema

CCSD |X| S

16.5 orca.coords.units

Keyword	orca.coords.units
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

|X| S

16.6 orca.cpcm

Keyword	orca.cpcm
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

|X| S

16.7 orca.expert

Keyword	orca.expert
Kind	repeating-record
Type	Repeating record with columns: String
Default	Required / No default
Deprecated	No

Use verbatim expert setting for orca.

Schema

|X| S ...

|X| ...

16.8 orca.geom.optimizationquality

Keyword	orca.geom.optimizationquality
Kind	scalar
Type	String
Default	TIGHTOPT

Deprecated	No
------------	----

Schema

TIGHTOPT |X| S

16.9 orca.method.allowrhf

Keyword	orca.method.allowrhf
Kind	scalar
Type	Logical
Default	F
Deprecated	No

Schema

F |X| L

16.10 orca.method.amass

Keyword	orca.method.amass
Kind	scalar
Type	String
Default	Mass2016
Deprecated	No

Schema

Mass2016 |X| S

16.11 orca.method.frozenscore

Keyword	orca.method.frozenscore
Kind	scalar
Type	Logical
Default	F
Deprecated	No

Schema

F |X| L

16.12 orca.method.grid

Keyword	orca.method.grid
Kind	scalar
Type	String
Default	GRID4
Deprecated	No

Schema

GRID4 |X| S

16.13 orca.method.gridx

Keyword	orca.method.gridx
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

|X| S

16.14 orca.method.method

Keyword	orca.method.method
Kind	scalar
Type	String
Default	B3LYP
Deprecated	No

Schema

B3LYP |X| S

16.15 orca.method.ri

Keyword	orca.method.ri
Kind	scalar
Type	Logical
Default	F
Deprecated	No

Schema

F |X| L

16.16 orca.method.runtype

Keyword	orca.method.runtype
Kind	scalar
Type	String
Default	OPT
Deprecated	No

Schema

OPT |X| S

16.17 orca.method.usesymm

Keyword	orca.method.usesymm
Kind	scalar
Type	Logical
Default	F
Deprecated	No

Schema

F |X| L

16.18 orca.mp2

Keyword	orca.mp2
Kind	scalar
Type	Logical
Default	F
Deprecated	No

Schema

F |X| L

16.19 orca.mp2type

Keyword	orca.mp2type
Kind	scalar
Type	String
Default	MP2
Deprecated	No

Schema

MP2 |X| S

16.20 orca.output.pdbfile

Keyword	orca.output.pdbfile
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

Schema

|X| L

16.21 orca.output.print

Keyword	orca.output.print
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

|X| S

16.22 orca.output.printlevel

Keyword	orca.output.printlevel
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

|X| S

16.23 orca.output.xyzfile

Keyword	orca.output.xyzfile
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

Schema

|X| L

16.24 orca.rel.method

Keyword	orca.rel.method
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

|X| S

16.25 orca.rel.soctype

Keyword	orca.rel.soctype
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

|X| S

16.26 orca.scf.autostart

Keyword	orca.scf.autostart
Kind	scalar
Type	Logical
Default	T
Deprecated	No

Schema

T |X| L

16.27 orca.scf.cnvdamp

Keyword	orca.scf.cnvdamp
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

Schema

|X| L

16.28 orca.scf.cnvshift

Keyword	orca.scf.cnvshift
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

Schema

|X| L

16.29 orca.scf.convergence

Keyword	orca.scf.convergence
Kind	scalar
Type	String
Default	NORMALSCF
Deprecated	No

Schema

NORMALSCF |X| S

16.30 orca.scf.convergencestrategy

Keyword	orca.scf.convergencestrategy
Kind	scalar
Type	String
Default	NormalConv
Deprecated	No

Schema

NormalConv |X| S

16.31 orca.scf.diis

Keyword	orca.scf.diis
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

Schema

|X| L

16.32 orca.scf.fracocc

Keyword	orca.scf.fracocc
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

|X| S

16.33 orca.scf.guess

Keyword	orca.scf.guess
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

|X| S

16.34 orca.scf.hftyp

Keyword	orca.scf.hftyp
Kind	scalar
Type	String
Default	RHF
Deprecated	No

Schema

RHF |X| S

16.35 orca.scf.jmatrix

Keyword	orca.scf.jmatrix
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

|X| S

16.36 orca.scf.kdiis

Keyword	orca.scf.kdiis
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

Schema

|X| L

16.37 orca.scf.keepdens

Keyword	orca.scf.keepdens
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

Schema

|X| L

16.38 orca.scf.keepints

Keyword	orca.scf.keepints
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

Schema

|X| L

16.39 orca.scf.kmatrix

Keyword	orca.scf.kmatrix
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

|X| S

16.40 orca.scf.maxiter

Keyword	orca.scf.maxiter
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

|X| S

16.41 orca.scf.nr

Keyword	orca.scf.nr
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

Schema

|X| L

16.42 orca.scf.readints

Keyword	orca.scf.readints
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

Schema

|X| L

16.43 orca.scf.scfmode

Keyword	orca.scf.scfmode
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

|X| S

16.44 orca.scf.smeartemp

Keyword	orca.scf.smeartemp
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

Schema

|X| L

16.45 orca.scf.soscf

Keyword	orca.scf.soscf
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

Schema

|X| L

16.46 orca.scf.uno

Keyword	orca.scf.uno
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

Schema

|X| L

16.47 orca.scf.usecheapints

Keyword	orca.scf.usecheapints
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

Schema

|X| L

16.48 orca.scf.valformat

Keyword	orca.scf.valformat
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

|X| S

16.49 orca.tole

Keyword	orca.tole
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Schema

|X| F

16.50 orca.tolmaxg

Keyword	orca.tolmaxg
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Schema

|X| F

16.51 orca.tolrmaxd

Keyword	orca.tolrmaxd
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Schema

|X| F

16.52 orca.tolrmsd

Keyword	orca.tolrmsd
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Schema

|X| F

16.53 orca.tolrmsg

Keyword	orca.tolrmsg
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Schema

|X| F

17 SIESTA

17.1 phsf.broadening

Keyword	phsf.broadening
Kind	scalar
Type	Float
Default	0.5
Deprecated	No

Schema

0.5 |X| F

17.2 phsf.ekeqp

Keyword	phsf.ekeqp
Kind	vector
Type	Vector: Float, Float
Default	0.0 0.0
Deprecated	No

Schema

0.0 0.0 |X| F F

17.3 phsf.numtimepoints

Keyword	phsf.numtimepoints
Kind	scalar
Type	Integer
Default	40000
Deprecated	No

Schema

40000 |X| I

17.4 siesta.AtomicCoordinatesFormat

Keyword	siesta.AtomicCoordinatesFormat
Kind	scalar
Type	String
Default	Fractional
Deprecated	No

Specify units and format of atomic coordinates. Borh - cartesian in Bohr. Ang - cartesian in Angstrom. ScaledCartesian - cartesian scaled by lattice constant. Fractional - coordinates are given in units of the lattice vectors.

Schema

Fractional |X| S

17.5 siesta.Beta.Broadening

Keyword	siesta.Beta.Broadening
Kind	scalar
Type	Float
Default	0.1
Deprecated	No

Schema

0.1 |X| F

17.6 siesta.Block.AtomicCoordinatesAndAtomicSpecies

Keyword	siesta.Block.AtomicCoordinatesAndAtomicSpecies
Kind	repeating-record
Type	Repeating record with columns: Float, Float, Float, Integer
Default	Required / No default
Deprecated	No

Specify coordinates of each atom as well as the species index. x1 y1 z1 SpeciesIndex1 x2 y2 z2 SpeciesIndex2 ...

Schema

```
|X| F F F I
|X| ...
```

17.7 siesta.Block.ChemicalSpeciesLabel

Keyword	siesta.Block.ChemicalSpeciesLabel
Kind	vector
Type	Vector: Integer, Integer, String
Default	Required / No default
Deprecated	No

Indicate the different chemical species in the unit cell. Format: SpeciesIndex1 AtomicNumber1 AtomicSymbol1 SpeciesIndex2 AtomicNumber2 AtomicSymbol2 ...

Schema

```
|X| I I S
```

17.8 siesta.Block.LatticeParameters

Keyword	siesta.Block.LatticeParameters
Kind	vector
Type	Vector: Float, Float, Float, Float, Float, Float
Default	Required / No default
Deprecated	No

Set lattice parameters, i.e., a b c alpha beta gamma

Schema

```
|X| F F F F F F
```

17.9 siesta.Block.PAO.Basis

Keyword	siesta.Block.PAO.Basis
Kind	repeating-record
Type	Repeating record with columns: String
Default	Required / No default
Deprecated	No

Set siesta basis manually

Schema

|X| S ...
|X| ...

17.10 siesta.Coreresponse.Broadening

Keyword	siesta.Coreresponse.Broadening
Kind	scalar
Type	Float
Default	0.5
Deprecated	No

Schema

0.5 |X| F

17.11 siesta.Diag.DivideAndConquer

Keyword	siesta.Diag.DivideAndConquer
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

|X| S

17.12 siesta.DM.InitSpin.AF

Keyword	siesta.DM.InitSpin.AF
Kind	scalar
Type	Logical
Default	Required / No default

Deprecated	No
------------	----

Set magnetic behavior of system to antiferromagnetic if true. Default is false and will be set up as ferromagnetic.

Schema

|X| L

17.13 siesta.DM.MixingWeight

Keyword	siesta.DM.MixingWeight
Kind	scalar
Type	Float
Default	0.10
Deprecated	No

Mixing weight for the density matrix.

Schema

0.10 |X| F

17.14 siesta.DM.NumberBroyden

Keyword	siesta.DM.NumberBroyden
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Set number of broyden steps

Schema

|X| I

17.15 siesta.DM.NumberBroyden

Keyword	siesta.DM.NumberBroyden
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Schema

|X| I

17.16 siesta.DM.NumberPulay

Keyword	siesta.DM.NumberPulay
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Set number of Pulay steps

Schema

|X| I

17.17 siesta.DM.Tolerance

Keyword	siesta.DM.Tolerance
Kind	scalar
Type	Float
Default	1.0e-4
Deprecated	No

Convergence tolerance for elements of the density matrix

Schema

1.0e-4 |X| F

17.18 siesta.ElectronicTemperature

Keyword	siesta.ElectronicTemperature
Kind	vector
Type	Vector: Float, String
Default	1.0 K
Deprecated	No

Set electronic teperature used to define occupation numbers during SCF.

Schema

1.0 K |X| F S

17.19 siesta.expert

Keyword	siesta.expert
Kind	repeating-record
Type	Repeating record with columns: String
Default	Required / No default
Deprecated	No

Schema

|X| S ...
|X| ...

17.20 siesta.LatticeConstant

Keyword	siesta.LatticeConstant
Kind	vector
Type	Vector: Float, String
Default	Required / No default
Deprecated	No

Set an overall multiplicative constant for the lattice parameters.

Schema

|X| F S

17.21 siesta.LongOutput

Keyword	siesta.LongOutput
Kind	scalar
Type	Logical
Default	T
Deprecated	No

Output lots of info.

Schema

T |X| L

17.22 siesta.MaxSCFIterations

Keyword	siesta.MaxSCFIterations
Kind	scalar
Type	Integer

Default	100
Deprecated	No

Max number of SCF iterations allowed.

Schema

100 |X| I

17.23 siesta.MeshCutoff

Keyword	siesta.MeshCutoff
Kind	vector
Type	Vector: Float, String
Default	120.0 Ry
Deprecated	No

Energy that defines the real-space grid. Point spacing scales as $1/\sqrt{(E_c)}$.

Schema

120.0 Ry |X| F S

17.24 siesta.NumberOfAtoms

Keyword	siesta.NumberOfAtoms
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Indicate number of atoms in unit cell.

Schema

|X| I

17.25 siesta.NumberOfSpecies

Keyword	siesta.NumberOfSpecies
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Indicate number of species in unit cell.

Schema

|X| I

17.26 siesta.PAO.BasisSize

Keyword	siesta.PAO.BasisSize
Kind	scalar
Type	String
Default	DZP
Deprecated	No

Define basis size. Can be SZ, DZ, SZP, DZP.

Schema

DZP |X| S

17.27 siesta.PAO.BasisType

Keyword	siesta.PAO.BasisType
Kind	scalar
Type	String
Default	split
Deprecated	No

Define basis type split: Split-valence scheme for multiple-zeta. The split is based on different radii. splitgauss: Same as split but using Gaussian functions. nodes: Generalized PAO's. nonodes: The original PAO's are used, multiple-zeta is generated by changing the scale-factors, instead of using the excited orbitals. filteret: Use the filterets as a systematic basis set. The size of the basis set is controlled by the filter cut-off for the orbitals.

Schema

split |X| S

17.28 siesta.SolutionMethod

Keyword	siesta.SolutionMethod
Kind	scalar
Type	String
Default	diagon
Deprecated	No

Method for solving the eigenvalue problem.

Schema

diagon |X| S

17.29 siesta.SpinPolarized

Keyword	siesta.SpinPolarized
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Include spin in calculation. Value should be yes or no.

Schema

|X| S

17.30 siesta.TD.CoreExcitedAtom

Keyword	siesta.TD.CoreExcitedAtom
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Set which atom will be the absorbing atom, i.e., which atoms the core-hole will be centered on

Schema

|X| I

17.31 siesta.TD.CorePerturbationCharge

Keyword	siesta.TD.CorePerturbationCharge
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Set the core-hole charge.

Schema

|X| F

17.32 siesta.TD.mxPC

Keyword	siesta.TD.mxPC
Kind	scalar
Type	Integer
Default	2
Deprecated	No

Maximum number of points to use for predictor-corrector algorithm.

Schema

2 |X| I

17.33 siesta.TD.NumberOfTimeSteps

Keyword	siesta.TD.NumberOfTimeSteps
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Specify number of time steps for RT-TDDFT run.

Schema

|X| I

17.34 siesta.TD.ShapeOfEfield

Keyword	siesta.TD.ShapeOfEfield
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Specify the shape of the field for the RT-TDDFT run. Options are: step - Turn off dipole field at time 0. halfstep - Dipole field changes from full strength to 1/2 strength at time 0. delta - Dipole field with time dependence delta(t). const - Turn on dipole field at time 0. sine - Dipole field with damped sine wave time dependence, with optional Gaussian envelope to create quasi-harmonic wave packet. core - turn on core-hole potential at time 0. core2 - GS with full core-hole. Excited states from linear response after.

Schema

|X| S

17.35 siesta.TD.TimeStep

Keyword	siesta.TD.TimeStep
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Specify the time step for RT-TDDFT run.

Schema

|X| F

17.36 siesta.UseSaveData

Keyword	siesta.UseSaveData
Kind	scalar
Type	Logical
Default	F
Deprecated	No

Use saved data to start siesta calculation

Schema

F |X| L

17.37 siesta.XC.authors

Keyword	siesta.XC.authors
Kind	scalar
Type	String
Default	PBE
Deprecated	No

Set authors of exchange correlation functional.

Schema

PBE |X| S

17.38 siesta.XC.functional

Keyword	siesta.XC.functional
Kind	scalar
Type	String
Default	GGA
Deprecated	No

Set flavor of exchange-correlation functional, i.e., LDA or GGA.

Schema

GGA |X| S

18 cif2cell

18.1 cif2cell.atomicunits

Keyword	cif2cell.atomicunits
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

print coordinates in bohr rather than angstroms

Schema

|X| L

18.2 cif2cell.cartesian

Keyword	cif2cell.cartesian
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

make output in cartesian form

Schema

|X| L

18.3 cif2cell.cif_input

Keyword	cif2cell.cif_input
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Crystallographic information file to use.

Schema

|X| S

18.4 cif2cell.outfile

Keyword	cif2cell.outfile
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Write output to outfile

Schema

|X| S

18.5 cif2cell.program

Keyword	cif2cell.program
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

```
% Program for cif2cell to write input for.  
|X| S
```

18.6 cif2cell.supercell

Keyword	cif2cell.supercell
Kind	vector

Type	Vector: Integer, Integer, Integer
Default	Required / No default
Deprecated	No

Schema

```
|X| I I I
```

19 Loop Workflows

19.1 loop_labels

Keyword	loop_labels
Kind	repeating-record
Type	Repeating record with columns: String
Default	Required / No default
Deprecated	No

define the labels that will be used for the loop directories. There should be one label for each value of the loop parameter.

Schema

```
|X| S
|X| ...
```

19.2 loop_parameter

Keyword	loop_parameter
Kind	repeating-record
Type	Repeating record with columns: String
Default	Required / No default
Deprecated	No

Define loop over a set of parameter values for a certain calculation. First line is the keyword to loop over. The rest of the lines are the values associated with each iteration of the loop, with one line per set of values. Note: This only works with single line keywords.

Schema

```
|X| S
|X| S ...
|X| ...
```

19.3 loop_target

Keyword	loop_target
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Set the target property to calculate during the loop.

Schema

|X| S

20 Configurational Averaging

20.1 cfavg_choose_random_absorbers

Keyword	cfavg_choose_random_absorbers
Kind	scalar
Type	Logical
Default	False
Deprecated	No

Schema

False |X| L

20.2 cfavg_egrid

Keyword	cfavg_egrid
Kind	scalar
Type	String
Default	"first"
Deprecated	No

Choose energy grid type for configurational averaging of spectra. "regular" would be regular in energy, while "first" uses the grid of the first calculation.

Schema

"first" |X| S

20.3 cfavg_max_configurations

Keyword	cfavg_max_configurations
Kind	scalar
Type	Integer
Default	10
Deprecated	No

Set the number of configurations used in disordered systems.

Schema

10 |X| I

20.4 cfavg_max_configurations

Keyword	cfavg_max_configurations
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Schema

|X| I

20.5 cfavg_target

Keyword	cfavg_target
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

|X| S ...

21 Fitting

21.1 fit.bond

Keyword	fit.bond
Kind	repeating-record
Type	Repeating record with columns: Integer
Default	Required / No default
Deprecated	No

Schema

|X| I ...
|X| ...

21.2 fit.datafile

Keyword	fit.datafile
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

|X| S

21.3 fit.parameters

Keyword	fit.parameters
Kind	repeating-record
Type	Repeating record with columns: String, Float
Default	Required / No default
Deprecated	No

Schema

|X| S F
|X| ...

21.4 fit.target

Keyword	fit.target
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Schema

|X| S

22 OCEAN

22.1 ocean.abpad

Keyword	ocean.abpad
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

For the calculation of the final state wavefunctions, the convergence of the highest bands can be very slow. The calculation can be sped up by adding some throwaway bands at the top. These bands are not considered when checking for convergence. ocean.abpad adds bands to the calculation. Only used for abinit.

Schema

|X| I

22.2 ocean.acell

Keyword	ocean.acell
Kind	vector
Type	Vector: Float, Float, Float
Default	Required / No default
Deprecated	No

Scaling in Bohr for the primitive vectors of the unit cell

Schema

|X| F F F

22.3 ocean.cnbse.broaden

Keyword	ocean.cnbse.broaden
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Half? width of broadening in eV for the spectrum. Must be larger than zero. Suggested value is the core-hole broadening.

Schema

|X| F

22.4 ocean.cnbse.gmres.elist

Keyword	ocean.cnbse.gmres.elist
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

List of energies at which to run the GMRES algorithm in eV. Lowest unoccupied state is set at zero without the core-hole.

Schema

|X| F ...

22.5 ocean.cnbse.gmres.erange

Keyword	ocean.cnbse.gmres.erange
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Use energies at with even spacing specified by emin emax estep

Schema

|X| F

22.6 ocean.cnbse.gmres.ffff

Keyword	ocean.cnbse.gmres.ffff
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Sets convergence criterion for GMRES.

Schema

|X| F

22.7 ocean.cnbse.gmres.gprc

Keyword	ocean.cnbse.gmres.gprc
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Set Lorentzian broadening in Hartree in the GMRES algorithm when pre-conditioning. Default is 0.5, which should be reasonable for K edges.

Schema

|X| F

22.8 ocean.cnbse.gmres.nloop

Keyword	ocean.cnbse.gmres.nloop
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Restart GMRES algorithm will restart if the subspace grows to size nloop.

Schema

|X| I

22.9 ocean.cnbse.mode

Keyword	ocean.cnbse.mode
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Which spectroscopy to calculate: "XAS", "XES" NRIXS and XRS are treated the same as XAS.

Schema

|X| S

22.10 ocean.cnbse.niter

Keyword	ocean.cnbse.niter
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Number of Haydock iterations. Default is 100.

Schema

|X| I

22.11 ocean.cnbse.rad

Keyword	ocean.cnbse.rad
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

One of the screening radius as defined by ocean.screen.shells that will be used in the BSE calculation.

Schema

|X| F

22.12 ocean.cnbse.solver

Keyword	ocean.cnbse.solver
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Choose between Haydock algorithm (full spectrum, no excitons) or GMRES (single energy, exciton density).

Schema

|X| S

22.13 ocean.cnbse.spect_range

Keyword	ocean.cnbse.spect_range
Kind	vector
Type	Vector: Integer, Float, Float
Default	Required / No default
Deprecated	No

Sets energies over which the spectrum will be calculated. nsteps emin emax

Schema

|X| I F F

22.14 ocean.cnbse.strength

Keyword	ocean.cnbse.strength
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Sets the strength for the two interaction terms in the BSE Hamiltonian. Default is 1, and for emission 0. This can be changed for analysis.

Schema

|X| F

22.15 ocean.cnbse.xmesh

Keyword	ocean.cnbse.xmesh
Kind	vector
Type	Vector: Integer, Integer, Integer
Default	Required / No default
Deprecated	No

Wave-functions are converted into the NIST BSE format and condensed onto a grid of size ocean.cnbse.xmesh. Default is guessed in the code.

Schema

|X| I I I

22.16 ocean.core_offset

Keyword	ocean.core_offset
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

If true, core-level shift will be calculated, false, no shift, or a number to specify the shift by hand.

Schema

|X| S

22.17 ocean.degauss

Keyword	ocean.degauss
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Broadening for the smearing function in Ryd. for metallic occupations in the DFT calculations.

Schema

|X| F

22.18 ocean.dft

Keyword	ocean.dft
Kind	scalar
Type	String
Default	abi
Deprecated	No

Which DFT code to use: qe - QuantumEspresso, abi - ABINIT

Schema

abi |X| S

22.19 ocean.dft_energy_range

Keyword	ocean.dft_energy_range
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Instead of setting the number of bands (ocean.nbands) the user may request an energy range in eV for the final state wave-functions. This is an estimate using the volume of the unit cell and may be unreliable. Default is 25 eV.

Schema

|X| F

22.20 ocean.diemac

Keyword	ocean.diemac
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Macroscopic dielectric matrix.

Schema

|X| F

22.21 ocean.ecut

Keyword	ocean.ecut
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Plane wave basis truncation energy in Rydberg

Schema

|X| F

22.22 ocean.edges

Keyword	ocean.edges
Kind	repeating-record
Type	Repeating record with columns: Integer, Integer, Integer
Default	Required / No default
Deprecated	No

Specify the atoms and edges to calculate. Each edge entry consists of 3 integers. The first, if greater than zero, is the index of the atom for which the edge should be calculated. The second and third numbers are the principle quantum number and angular momentum. '1 2 1' denotes the L edges of the first atom. If the first number is negative, then it is interpreted as -Z, where Z is the atomic number, and in that case edges of all atoms with that atomic number will be calculated. '-22 2 1' will run the L edges of every titanium atom in the cell.

Schema

```
|X| I I I
|X| ...
```

22.23 ocean.fband

Keyword	ocean.fband
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Determines the number of occupied bands to be included in the SCF calculation of the density. For insulators the default (fband = 0.125) is fine, but for metals the highest band in the density calculation should have no occupation weight. Using fband, the number of bands is determined by $n = n_{\text{atom}} * \text{fband}$.

Schema

```
|X| F
```

22.24 ocean.k0

Keyword	ocean.k0
Kind	vector
Type	Vector: Float, Float, Float
Default	Required / No default
Deprecated	No

DFT states are calculated using a shifted k-point grid. The first k-point is given by: $1/(N_k(1) * k0(1))$, $1/(N_k(2) * k0(2))$, $1/(N_k(3) * k0(3))$ where N_k is given by ocean.nkpt. The default shift is $k0 = 1/8, 2/8, 3/8$.

Schema

|X| F F F

22.25 ocean.ldau

Keyword	ocean.ldau
Kind	repeating-record
Type	Repeating record with columns: Logical
Default	Required / No default
Deprecated	No

Controls the U parameters for the DFT run. Only works with QuantumEspresso currently. Not sure what the parameters are, but I believe they are logical flag to run/not run LDA+U, U for each atom that has spin associated with smag.

Schema

|X| L
|X| F
|X| ...

22.26 ocean.metal

Keyword	ocean.metal
Kind	scalar
Type	Logical
Default	Required / No default
Deprecated	No

If True, the code will complain unlessoccopt is 3. If False, the codes expectsoccopt = 1.

Schema

|X| L

22.27 ocean.natom

Keyword	ocean.natom
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Number of atoms in the unit cell

Schema

|X| I

22.28 ocean.nbands

Keyword	ocean.nbands
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Number of bands for the final state calculations. This includes valence and conduction bands.

Schema

|X| I

22.29 ocean.ngkpt

Keyword	ocean.ngkpt
Kind	vector
Type	Vector: Integer, Integer, Integer
Default	Required / No default
Deprecated	No

Number of k-points in each direction used to sample the cell for the ground-state calculation of the density.

Schema

|X| I I I

22.30 ocean.nkpt

Keyword	ocean.nkpt
Kind	vector
Type	Vector: Integer, Integer, Integer
Default	Required / No default
Deprecated	No

Number of k-points in each direction used to sample the cell for the calculation of the final states.

Schema

|X| I I I

22.31 ocean.nspin

Keyword	ocean.nspin
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Choose paramagnetic (nspin = 1), or spin dependent (nspin = 2).

Schema

|X| I

22.32 ocean.nstep

Keyword	ocean.nstep
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Maximum number of iterations for the DFT SCF calculations. Default is 50

Schema

|X| I

22.33 ocean.ntypat

Keyword	ocean.ntypat
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Number of different types (usually species) of atom in the unit cell.

Schema

|X| I

22.34 ocean.occpt

Keyword	ocean.occpt
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Controls determination of occupation. Most important values are: 1 - States are all doubly degenerate and either occupied or empty depending on band. 3 - States are all doubly degenerate but can have fractional occupations depending on Fermi function. Suitable for metals.

Schema

|X| I

22.35 ocean.opf.fill

Keyword	ocean.opf.fill
Kind	vector
Type	Vector: Integer, String
Default	Required / No default
Deprecated	No

atomic number followed by the name of the .fill file to use for this calculation.

Schema

|X| I S

22.36 ocean.opf.hfkgrid

Keyword	ocean.opf.hfkgrid
Kind	vector
Type	Vector: Integer, Integer
Default	Required / No default
Deprecated	No

The first parameter sets the grid for hfk.x, and should be left at 2000. The second determines the maximum number of projectors per angular momentum channel. The default of 20 will work for many calculations.

Schema

|X| I I

22.37 ocean.opf.opts

Keyword	ocean.opf.opts
Kind	vector
Type	Vector: Integer, String
Default	Required / No default
Deprecated	No

atomic number followed by the name of the .opts file to use for this calculation.

Schema

|X| I S

22.38 ocean.opf.program

Keyword	ocean.opf.program
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Use oncvpsp UPS pseudopotentials with qe

Schema

|X| S

22.39 ocean.para_prefix

Keyword	ocean.para_prefix
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Set the command to use for parallel runs, i.e., mpirun, mpiexec etc, along with any flags to use.

Schema

|X| S ...

22.40 ocean.photon.energy

Keyword	ocean.photon.energy
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Set energy of x-rays at edge

Schema

|X| F

22.41 ocean.photon.operator

Keyword	ocean.photon.operator
Kind	scalar
Type	String
Default	dipole
Deprecated	No

Select the operator to be used to calculate the (dipole or quad)

Schema

dipole |X| S

22.42 ocean.photon.polarization

Keyword	ocean.photon.polarization
Kind	vector
Type	Vector: Float, Float, Float
Default	Required / No default
Deprecated	No

Direction of polarization for XANES or XES calculations

Schema

|X| F F F

22.43 ocean.photon.qhat

Keyword	ocean.photon.qhat
Kind	vector
Type	Vector: Float, Float, Float
Default	Required / No default
Deprecated	No

Direction of q-vector for quadrupole or NRIXS calculations

Schema

|X| F F F

22.44 ocean.photon_q

Keyword	ocean.photon_q
Kind	vector
Type	Vector: Float, Float, Float
Default	Required / No default
Deprecated	No

For UV/VIS and RIXS calculations, the valence and conduction band states must be offset from each other (by the momentum that is absorbed or transferred). This sets that momentum offset in units of the reciprocal lattice vectors of the system. Default is 0 0 0.

Schema

|X| F F F

22.45 ocean.pp_list

Keyword	ocean.pp_list
Kind	repeating-record
Type	Repeating record with columns: String
Default	Required / No default
Deprecated	No

Names of the pseudopotential files to be used listed in the same order as ocean.znucl

Schema

|X| S
|X| ...

22.46 ocean.rprim

Keyword	ocean.rprim
Kind	matrix
Type	3 rows of Float, Float, Float
Default	Required / No default
Deprecated	No

Primitive vectors of unit cell

Schema

```
|X| F F F  
|X| F F F  
|X| F F F
```

22.47 ocean.scfac

Keyword	ocean.scfac
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

The is the Slater integral factor to scale the slater integrals. For 3d transition metals 0.8 is a reasonable value, while 0.6 is better for f-electron atoms.

Schema

```
|X| F
```

22.48 ocean.scratch

Keyword	ocean.scratch
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Where to write scratch files

Schema

```
|X| S
```

22.49 ocean.screen.nbands

Keyword	ocean.screen.nbands
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Number of bands to use for the screening calculations. This requires a large number of bands. Should be about 100eV above the Fermi level, but convergence should be checked.

Schema

|X| I

22.50 ocean.screen.nkpt

Keyword	ocean.screen.nkpt
Kind	vector
Type	Vector: Integer, Integer, Integer
Default	Required / No default
Deprecated	No

Number of k-point to be used for the screening calculations. Default is 2 2 2, and is sufficient for a wide variety of systems, although very small unit cells might require more, and large unit cells may only require gamma point.

Schema

|X| I I I

22.51 ocean.screen.shells

Keyword	ocean.screen.shells
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

The screening calculation is RPA at small radius and a model at large radius. The cross-over between the RPA and model is set by shells in Bohr. Several different radii can be chosen to look at the convergence. Convergence is usually reached at around 3-4 Bohr. The supercell defined by ocean.paw.nkpt must have dimensions larger than the radius specified by ocean.screen.shells or results may be unreliable. The default of 3.5 Bohr is usually OK. Values > 6 Bohr should not be used.

Schema

|X| F ...

22.52 ocean.screen_energy_range

Keyword	ocean.screen_energy_range
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Instead of setting number of bands for the screening, the user may request an energy range in eV. This is only an estimate, and may need to be converged.

Schema

|X| F

22.53 ocean.ser_prefix

Keyword	ocean.ser_prefix
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

prefix for serial parts of the code, for example, cut3d should be run with only 1 processor, i.e.,
mpirun -n 1

Schema

|X| S ...

22.54 ocean.smag

Keyword	ocean.smag
Kind	repeating-record
Type	Repeating record with columns: Integer, Integer, Integer
Default	Required / No default
Deprecated	No

Controls initial magnetism for the DFT calculations.

Schema

|X| I I I

|X| ...

22.55 ocean.smag.ixc

Keyword	ocean.smag.ixc
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Used for abinit runs in smag card. Don't know the meaning at the moment. This should be combined with ocean.smag to write the card to the ocean input.

Schema

|X| I

22.56 ocean.spin-orbit

Keyword	ocean.spin-orbit
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

If ocean.spin-orbit ≥ 0 , the code will not automatically calculate the spin-orbit splitting.

Schema

|X| F

22.57 ocean.toldfe

Keyword	ocean.toldfe
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Total energy convergence parameter for the density run. Default is 10^{-6}

Schema

|X| F

22.58 ocean.tolwfr

Keyword	ocean.tolwfr
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Convergence criterium for the non-scf wave-function calculations. Default is 10^{-16}

Schema

|X| F

22.59 ocean.typat

Keyword	ocean.typat
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

Type of each atom in the unit cell denoted by the integer index corresponding to the ocean.znucl list of atomic numbers. This list must be in the same as the coordinates listed in ocean.xred

Schema

|X| I ...

22.60 ocean.xred

Keyword	ocean.xred
Kind	repeating-record
Type	Repeating record with columns: Float, Float, Float
Default	Required / No default
Deprecated	No

Reduced coordinats of atoms in the unit cell.

Schema

|X| F F F
|X| ...

22.61 ocean.znucl

Keyword	ocean.znucl
Kind	scalar
Type	Integer
Default	Required / No default
Deprecated	No

List of length ntypat with atomic numbers of each type of atom in the unit cell

Schema

|X| I ...

22.62 ocean.zymb

Keyword	ocean.zymb
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

For use with QE. Each element listed in ocean.znucle can have a symbol associated with it.

Schema

|X| S ...

23 Miscellaneous

23.1 ene_int

Keyword	ene_int
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Internal (electronic) energy of the system, in au.

Schema

|X| F

23.2 mac_diel_const

Keyword	mac_diel_const
Kind	scalar
Type	Float
Default	Required / No default
Deprecated	No

Approximate value of the macroscopic dielectric constant of the system used in some methods to accelerate convergence of the SCF cycle.

Schema

|X| F

23.3 magmom_by_label

Keyword	magmom_by_label
Kind	repeating-record
Type	Repeating record with columns: String, Float
Default	Required / No default
Deprecated	No

set spin for elements of a cif file by the label

Schema

|X| S F

|X| ...

23.4 mp_apikey

Keyword	mp_apikey
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Specify your mp API key. This should probably be done with an environment variable instead for security.

Schema

|X| S

23.5 mp_id

Keyword	mp_id
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

Single material project id. Can be mp-NNNN or just the number.

Schema

|X| S

23.6 spectralFunction_file

Keyword	spectralFunction_file
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

file to read spectral function from - 2 column file

Schema

|X| S

23.7 xanes_file

Keyword	xanes_file
Kind	scalar
Type	String
Default	Required / No default
Deprecated	No

file to read xanes from - 2 column file

Schema

|X| S